Marketing Analysis for As Leiteiras

Table of Contents

1. Introduction

1.1 Project background

As Leiteiras began as an idea for distributing fresh organic milk in returnable glass bottles around Lugo. It is still a proposed business rather than an operating one: there is no delivery van, established route, employee or customer contract. Omar who is the project owner later selected Sen Máis as a real supplier reference because he wanted to test the idea using actual prices, supply conditions and operating constraints rather than continue working only with hypothetical figures. Sen Máis is therefore treated in this study as a potential supplier, not as a business partner already committed to the project.

1.2 Purpose and scope

The purpose of this thesis is to test whether the proposed delivery model could work under realistic conditions in Lugo. The analysis focuses on whether enough customers are likely to accept the proposed price, whether deliveries can be organised efficiently, and whether B2B customers could provide the volume that household sales alone may not generate. The competitive environment is also considered because customers can already buy similar products in Lugo.

Milk production itself is outside the scope of the project. Sen Máis, a CRAEGA-certified organic dairy farm near Palas de Rei, is treated as the potential supplier in the model. It would produce, process and bottle the milk, while As Leiteiras would be responsible for distribution and customer relationships. The thesis therefore concentrates mainly on demand, market positioning, delivery economics and their implications for financial viability.

1.3 Research questions

The thesis is organised around one main research question:

- Does it make financial sense to launch the As Leiteiras home-delivery model in Lugo?

This is divided into five thesis-level sub-questions (SQ1–SQ5):

- SQ1 — Break-even: What daily sales volume and selling price would be needed for the business to cover its costs?
- SQ2 — Staffing: At launch, would a full-time driver be financially justified, or would starting with a part-time driver reduce the risk?
- SQ3 — Customer channel: Which customer group offers better delivery economics for As Leiteiras: households or restaurants and cafés?
- SQ4 — Sensitivity: How much do the results change when the selling price, litres delivered per stop, or fuel cost changes?
- SQ5 — Replication: What conditions would be required for this model to be considered in another medium-sized Galician city?
- SQ1 (costs and break-even) and SQ2 (hiring) are primarily financial questions, addressed in the financial model. This marketing analysis addresses the demand side of SQ3 directly (B2C vs B2B segmentation) and provides the WTP, market-sizing, and geographic inputs used jointly with the financial model to answer SQ1, SQ4 and SQ5.

1.4 Summary of conclusions

Overall, the current results do not support launching the business with a dedicated refrigerated van and a full-time driver. Under the price and cost assumptions used in the model, this structure would not be profitable. However, this does not necessarily mean that the whole business idea should be rejected. B2C demand at €2.10/L appears limited, while B2B volume is necessary but is still not enough to make the current model viable. Section 6 discusses the reasons for this result and the additional information needed before any further investment.

1.5 Structure of the thesis

The rest of this chapter is structured as follows. Section 2 sets out the literature used in the analysis, while Section 3 explains the data and methods. Section 4 reports the main results, Section 5 interprets them, and Section 6 brings the analysis together in the final conclusions and recommendations.

2. Literature Review and Theoretical Framework

The literature review focuses on three issues that recur throughout the analysis: the gap between stated WTP and actual purchasing behaviour, the effect of delivery density on last-mile costs, and consumer responses to local and organic food.

2.1 Willingness to Pay and Hypothetical Bias

Willingness to pay surveys are widely used, but stated WTP does not always match actual purchasing behaviour. The Lugo questionnaire asked respondents to select a fixed price range and did not include a real purchase decision, a validation question or a cheap-talk script, so the figures it produced need to be interpreted as stated preferences rather than as observed behaviour. For this thesis I use 1.35 as the main reference point for the likely gap between stated and actual WTP, rather than a more aggressive multiplier: Murphy et al. (2005), reviewing 28 studies containing 83 observations, report this as the median hypothetical-to-actual ratio. Larger gaps did occur in some of the studies they reviewed, but they were not typical, and treating stated WTP as two or three times actual WTP would not be supported by that evidence.

The proposed offer was unfamiliar to most respondents, particularly because home delivery was almost absent from their current purchasing behaviour. Unfamiliarity of this kind is a reasonable concern, and it may well widen the gap between what people say and what they do. What it does not do is indicate by how much. Murphy et al. (2005) does not itself establish that unfamiliar products produce systematically higher hypothetical bias; that claim comes from later stated-preference research building on the meta-analysis. Cheap-talk scripts and incentive compatible mechanisms are associated with smaller bias in that broader literature (Harrison & Rutström, 2008), and the Lugo questionnaire uses neither, which is worth noting as a limitation even though this thesis does not attempt to quantify its effect. The available data are therefore not strong enough to place this survey confidently at the upper end of the bias distribution, and I treat 1.35 as a reference point rather than a precise correction factor. The demand analysis in Section 4.4 accordingly applies a separate conservative conversion assumption instead of mechanically dividing stated WTP by 1.35; that 20% figure is a planning assumption, not an estimate derived from the Lugo survey and not a market-specific or product-specific empirical benchmark.

2.2 Last-mile Food Distribution Economics

Delivery economics are particularly important for As Leiteiras because the model relies on one driver and one vehicle. The issue is not simply how many customers can be reached, but how closely those customers can be grouped on a route. Gevaers et al. (2011) identify stop density, geographic coverage and market penetration as important determinants of last-mile distribution costs. This is the reason the later analysis pays attention to the concentration of delivery stops rather than treating every additional customer as equally valuable. A group of customers in Lugo city, for example, can have very different route implications from the same number of customers spread across surrounding municipalities. B2B accounts are also relevant because more litres can potentially be delivered at each stop.

The proposed model avoids another source of operating complexity by leaving milk production, processing and bottling with Sen Máis. As Leiteiras would concentrate on collection, delivery and customer management rather than investing in dairy-processing equipment. This is broadly consistent with the asset-light approach discussed by Bourlakis and Weightman (2004). The trade-off is greater dependence on one supplier. Purchase price, available volume and product quality would all depend on Sen Máis, and there is currently no formal supply or exclusivity agreement between the two parties.

2.3 Consumer Behaviour for Short Food Supply Chains and Organic Products

Moser et al. (2011) is useful here mainly for explaining why attributes such as organic certification and local origin can affect consumer choice, rather than for providing a numerical premium for As Leiteiras. Their review covers credence attributes in fresh fruit and vegetables and finds that organic and local signals can be associated with higher willingness to pay. Because the evidence is not specific to dairy products, I do not transfer any of the reported premiums directly to organic milk in Lugo. This distinction matters in the present market: local origin and organic certification may add value, but similar attributes are already available through established Galician products. The literature therefore supports treating these characteristics as possible sources of differentiation, not as evidence that consumers in Lugo will automatically accept a higher price.

One possible explanation is that Lugo consumers already have many regional dairy products available at normal supermarket prices. Local origin is therefore not a new feature for them. Adding another local or organic label may create some extra value, but the survey does not show that most consumers are willing to pay a large premium for it.

As for As Leiteiras, the main issue is how the offer is positioned. Local milk is already common in Galicia, so local origin alone gives limited differentiation. The survey also provides little evidence that organic certification by itself can support the proposed price. Freshness, glass-bottle packaging and home delivery may therefore need to play a larger role in the value proposition. Short food supply chains are also relevant to the proposed model. Kneafsey et al. (2013) describe several forms, including farm sales, farmers' markets and delivery schemes. As Leiteiras is closest to the delivery model. In this thesis, I distinguish between two possible sources of value: proximity to the producer and convenience for the customer.

As Leiteiras could offer both. Sen Máis is located around 34 km from Lugo, while home delivery could save customers a separate shop visit. However, the survey suggests that local origin on its own is unlikely to justify the price premium. Freshness and delivery convenience may therefore be more useful elements of the positioning.

3. Methodology: Data Sources, Author Role, and Analytical Approach

3.1 Research Design, Data Sources and Author Role

This thesis adopts an external analyst perspective. The primary empirical materials used in the analysis were provided by the project founder, Omar. These include a consumer survey of 94 Lugo area respondents conducted in May–June 2025 and interview notes from an in depth conversation with Ramón Barreiro, co-founder of a comparable milk distribution initiative that operated in Santiago de Compostela between 2010 and 2016. The author did not participate in the original collection of these data. The author's contributions included reviewing and organising the data, conducting descriptive analysis and willingness-to-pay (WTP) segmentation, and constructing ordered logistic regression models. Further contributions included developing B2C and B2B sensitivity analysis scenarios and cross-validating the findings with secondary market data, publicly available demographic data, competitor pricing evidence, and academic literature. This analysis is an independent strategic assessment based on original data provided by the project initiator; the author did not conduct original field research.

3.2 Marketing Analysis Methods

Consumer demand is characterised through descriptive statistics and an ordered logistic regression applied to the survey data (Section 4.4), including a rule-based segmentation of respondents by stated purchase driver (Section 4.3).

To keep the demand estimates separate from the financial targets, I use three different terms. Market size refers to the population base before any WTP or conversion filter is applied. The population-based demand proxy applies the €2.00/L WTP threshold and a conservative conversion assumption to that population; it is an illustrative estimate rather than a sales forecast. Active customers refers to the accounts used in the financial model's stable-year scenario: 180 B2C households and 30 B2B accounts. I use €2.00/L as the WTP threshold because the survey did not test €2.10 directly; the €2.00–€2.50 bracket is the closest available category containing the proposed retail price.

For the competitor analysis, I compare products on five characteristics: freshness, packaging, certification, availability in Lugo and distribution channel.

3.3 Operations and Logistics Analysis

The route analysis uses the stable-year cost structure from the financial model, allowing different channel scenarios to be compared using EBITDA (Sections 4.2 and 4.7).

3.4 Financial Integration and Sensitivity Analysis

Section 4.7 combines B2C and B2B scenarios with the financial model, testing how demand, revenue, cost and profit move together under different conversion and pricing assumptions. Section 4.8 then checks these marketing-derived assumptions against the financial model's own inputs, to confirm the two are consistent.

3.5 Secondary Data and Triangulation

Secondary data (INE population statistics, retailer price observations, Arqueixal delivery route information, and the Banco de España macroeconomic projections used in the financial model) are used for background analysis and triangulation of key findings.

3.6 Methodological Limitations

The approach follows established guidance on applied management research (Gill & Johnson, 2010), with the author in an external analyst role rather than that of a participant observer. The absence of direct fieldwork access (the author did not administer the survey or conduct the Barreiro interview) is a recognised limitation. It is addressed throughout the analysis by being explicit about data provenance, distinguishing between empirically grounded and inference-based claims, and applying conservative adjustments when translating stated preferences into demand estimates. One exception is a small supplementary observation, conducted directly by the author: informal visits to Sen Máis retail stockists in Lugo, reported in the retail-channel section below, which is flagged there as the author's own non-systematic field check rather than part of the founder-supplied primary materials.

4. Results

4.1 Proposed Business Model and Potential Relationship with Sen Máis

As Leiteiras would operate as a last-mile distributor rather than a producer. The model assumes Sen Máis, a CRAEGA-certified organic dairy farm in A Cernada, Palas de Rei, as the potential supplier; no supply agreement has been signed. Sen Máis is approximately 34 km from Lugo by road, and for cost modelling, the financial model applies a more conservative 46 km one-way route assumption. The proposed business would deliver this milk to two client groups: households and HORECA establishments in the city.

As Leiteiras would purchase milk at the farm-gate at €1.65/L and sell it at €2.10/L (B2C) or €1.95/L (B2B/HORECA) under its own delivery and service brand. The physical product and bottle label would remain associated with Sen Máis unless a separate private label or co branding agreement were negotiated. Under this arrangement, Sen Máis would handle production, CRAEGA organic certification, pasteurisation, and bottling in returnable 1 litre glass bottles, while As Leiteiras would handle ordering, route planning, delivery, bottle collection, and customer relationships. Under this working hybrid brand architecture, Sen Máis remains the product brand displayed on the bottle, while As Leiteiras is the customer-facing delivery-service brand used for ordering, communications, vehicle identification and customer relationships; As Leiteiras would not initially introduce a separately labelled milk product.

The proposed service would distribute the same Sen Máis milk that is already available through some Lugo neighbourhood shops. According to Omar, the retail price is around €2.10/L, although this has not been independently confirmed. The model assumes that Sen Máis would continue supplying existing shops in Lugo and that As Leiteiras would have no exclusivity. Under this assumption, As Leiteiras would compete mainly through its delivery service rather than through a distinct milk product. No such commercial arrangement has yet been agreed with Sen Máis. Whether As Leiteiras presents itself as a delivery service for Sen Máis or as an independent retailer would affect pricing power, customer acquisition and long-term business resilience.

4.2 Market Geography: Serviceable Area, Municipality Analysis, and Route Feasibility

The survey cannot be used to compare demand across municipalities because 90.4% of respondents were from Lugo city. The service area is therefore assessed using population and transport data instead. Two factors are considered: the practical distance and driving time of each area, and the likely concentration of delivery stops. Sen Máis is around 34 km from Lugo, so the trip to the farm is treated separately from the customer delivery route.

Table 1 assesses the eight municipalities in the Lugo region based on population, distance, and density, and assigns a delivery level to each municipality. Route design scenarios (Table 2, Appendix B) and a detailed limitations discussion appear in Appendix B.

Table 1. Municipality assessment: Comarca de Lugo (INE 2024)

Municipality	Population (INE 2024)	Distance from Lugo	Density (inh./km2)	Delivery tier	Logistics note
Lugo (city)	99,482	—	302	Core	Primary delivery zone; urban density supports efficient B2C routing
Outeiro de Rei	5,358	13 km	40	Tier 1	A 6 motorway access; residential growth area; combinable with Rábade
Rábade	1,499	14 km	290	Tier 1	3 km from Outeiro de Rei; highest density of suburban municipalities
O Corgo	3,370	18 km	22	Tier 1	Relatively close to Lugo; rural roads; viability depends on B2B anchor accounts and customer clustering
Guntín	2,529	19 km	16	Tier 1	2 survey respondents; low density; route viable only with B2B stops
Friol	3,598	25 km	12	Tier 2	292 km2 area; very dispersed; not viable at launch without scale
Castroverde	2,497	28 km	14	Tier 2	Rural; not recommended for initial operations
Portomarin	1,286	42 km	9	Excluded	Beyond operational range; marginal population; excluded from model
Total (Core + Tier 1)	**112,238**	—	—	**Core + Tier 1**	**Launch area (Core) plus conditional near-term expansion area (Tier 1); Tier 1 is not part of the initial launch and depends on confirmed B2B anchor demand (see Appendix B.5).**

In terms of market size: it can be divided into three levels. The TAM covers people in Lugo and the surrounding comarca who drink cow's milk. The SAM is narrower and includes only customers who could realistically be served under the current supply, cold-chain and single-van constraints. In practice, this means Lugo city together with the Tier 1 municipalities. The SOM is smaller again and refers to the customer base that the business could realistically reach in the near term. In the financial model, the stable-year assumption is 180 B2C households and 30 B2B accounts.

The Core, Tier 1, Tier 2 and Excluded categories are used to show how the serviceable market differs in terms of route feasibility. Lugo city forms the Core area. Tier 1 covers nearby municipalities that could be considered if enough B2B anchor customers are secured. Tier 2 would require higher route density or additional delivery capacity before it became practical. Municipalities excluded are not within the scope of the operations used in this analysis.

Sources: IGE (Instituto Galego de Estatística) official municipal population table, 2024 column; road distances from distanciasentreciudades.com and Google Maps; density derived from INE surface area data. Lugo city figure of 99,482 is the official INE 2024 annual figure.

Lugo city contains 88.6% of the combined Core and Tier 1 population of 112,238. It is also much denser than the surrounding municipalities: 302 inhabitants per km², compared with 40 in Outeiro de Rei. Moving beyond the city therefore adds relatively little population while making the route more dispersed. O Corgo and Guntín illustrate this problem. They are about 18–19 km from Lugo, only four survey respondents came from these two municipalities, and neither is on the Sen Máis pickup route. Serving them would only make sense if enough customers could be grouped together, especially B2B accounts.

The figures in Section 4.4 point in the same direction. Adding the Tier 1 population increases the B2C demand proxy only slightly, so there is little evidence for expanding the household route outside Lugo at this stage. Lugo city remains the most practical area to test because it combines most of the reachable population with much higher density. This does not make the current business model profitable; it only identifies the least difficult geography. Expansion to Tier 1 or Tier 2 would need a separate commercial reason, most likely concentrated B2B demand.

4.3 Consumer Profile: Behavioural, Psychographic, and Sociodemographic Analysis

The consumer profile in this section comes from a survey of 94 people in Lugo, carried out by the project founder in May–June 2025. The survey asked about behaviour and personal values, but it did not ask about age, sex, family size, income or education level. Because of this, the survey cannot be used to build a sociodemographic profile of the respondents. Where demographic information is needed, this thesis uses INE data instead, only to give general context about Lugo. Appendix C gives full details of what the survey did and did not cover.

Of the 94 respondents, 81 (86%) said that they currently drink cow's milk. Nearly half of the respondents (47.9%) reported drinking 1–3 litres of milk per week. These are individual responses, however, because the questionnaire did not record household size. Buying habits were much more concentrated: 88% normally bought milk from supermarkets, compared with 6% using neighbourhood shops, while only one respondent used home delivery. Larsa, Feiraco and LEYMA were among the most frequently mentioned brands. This matters because "local product" was also the most common purchase factor in the survey, cited by 32% of respondents. However, familiarity with Galician dairy brands does not necessarily mean that these consumers would be willing to switch to a home-delivery service.

The purchase driver question (91 valid responses) was manually divided into three categories based on the primary drivers. This categorization method is descriptive rather than statistical. Table 4 summarises these categorization results.

Table 4. Consumer segments by primary purchase driver and residual status (n = 94; 91 valid purchase driver responses)

Segment	n (%)	Primary driver	WTP ≥ €2	WTP ≥ €1.50	Relevance for As Leiteiras
Quality & taste seekers	31 (33%)	Calidade / Sabor	2 (6%)	13 (42%)	Higher WTP than other segments; relevant for B2C testing
Local & eco advocates	38 (40%)	Local / Eco / Welfare	1 (3%)	8 (21%)	Largest group; lower WTP than quality/taste segment
Price driven buyers	19 (20%)	Prezo	0 (0%)	3 (16%)	Lowest WTP; low priority at the proposed price
Non drinkers / unclear	6 (6%)	Mixed	0	1	Minor; aspirational interest but no current consumption habit

This overview defines a narrow B2C opportunity; its size and financial impact are quantified in Sections 4.4 and 4.7. The main limitations are the lack of sociodemographic variables, the potential bias of the founder's convenience sample, and the analyst's discretion in categorizing open-text purchase driver responses into different groups; a full discussion of these limitations can be found in Appendix C.

4.4 Willingness to Pay: Survey Evidence, Interpretation, and Pricing Implications

The WTP data come from a survey of 94 respondents in Lugo, conducted by Omar in May–June 2025. Respondents were asked how much they would be willing to pay for one litre of locally produced organic milk in a returnable glass bottle. They selected one price range from a set of fixed options. The sample was collected by convenience, so the results here are exploratory rather than representative of the Lugo population. The results reported here are based on stated WTP. As discussed in Section 2.1, stated WTP may be higher than actual purchasing behaviour.

The survey included four price brackets, ranging from "less than €1.00" to "€2.00–€2.50". Table 6 shows the full distribution.

Table 6. Stated WTP distribution (n = 94)

WTP bracket	Respondents (n)	% of sample	Cumulative n (≥ bracket)	Cumulative %
Less than €1.00	10	10.6%	89	94.7%
€1.00 – €1.50	54	57.4%	79	84.0%
€1.50 – €2.00	22	23.4%	25	26.6%
€2.00 – €2.50	3	3.2%	3	3.2%
No answer	5	5.3%	—	—

Source: Lugo consumer survey, carried out by the project founder in May-June 2025. The cumulative columns show the number and share of respondents willing to pay at least the lower bound of each bracket. The denominator is the full sample (n = 94) in every row; the five respondents who did not answer are counted in the denominator but not counted toward any bracket's numerator.

The proposed B2C price of €2.10/L sits at the very top of the range tested in the survey. Most respondents were well below it: 57.4% selected €1.00–€1.50/L, and 26.6% reached at least €1.50/L. Only three respondents, or 3.2% of the sample, selected the €2.00–€2.50 bracket. This does not mean that all three would actually pay €2.10/L, because the questionnaire did not ask about that exact price. The 3.2% figure is therefore better read as an upper bound on possible acceptance of €2.10/L rather than as an acceptance rate for that price.

Looking beyond the overall price distribution, consumption volume adds another difficulty. The 3–5 L/week group, which would be attractive from a sales-volume perspective, showed the weakest willingness to pay: none reached €2.00/L and only 8% reached €1.50/L. By contrast, 41% of respondents drinking less than 1 L/week reached at least €1.50/L. In other words, the consumers who could contribute more litres per week were not the ones showing the strongest acceptance of a premium price. By primary purchase driver, quality- and taste-motivated buyers show the highest premium acceptance (40% and 45% ≥ €1.50/L respectively), while "local product", the largest single segment (30 respondents), shows only 23% at €1.50/L and 3% at €2.00/L. A positioning built solely on local provenance therefore reaches the largest interested group but not the group most willing to pay; quality- and taste-oriented respondents form a smaller group but show higher stated WTP. Full cross-tabulations (Tables 7 and 8, Appendix D) and Figure 2 (WTP distribution chart) appear in Appendix D.

For the reasons set out in Section 2.1, stated WTP is expected to overstate real purchasing behaviour here: the instrument used no validation or cheap-talk step, and the data show the gap directly: among the 83 respondents currently buying at a supermarket, only one stated WTP ≥ €2.00. The 20% conversion rate applied below is accordingly a deliberately conservative planning assumption, not an empirically derived figure; full instrument-design discussion appears in Appendix D.

Applying this to the Lugo city Core serviceable area (Section 4.2): 99,482 (Lugo city population, INE 2024) × 3.2% (stated WTP ≥ €2.00) × 20% (conversion) ≈ 637 demand units (population-based proxy). This is a demand proxy, not an active-customer forecast; the financial model uses a more conservative 180 active B2C households in the stable year. Even at full stable-year B2C volume (180 × 3 L/week × 52 weeks ≈ 28,080 L/year), the B2C channel alone does not cover the operating cost base, so additional B2B volume is required for route viability. Applying the same approach to the core plus first-level population (112,238) yields approximately 718 units, a slight increase of only about 81 units, confirming that B2C geographic expansion alone is unlikely to alter the demand situation.

Ordinal logistic regression analysis (Table 9) was performed on four WTP intervals to determine which respondent characteristics were associated with higher willingness to pay; since these intervals constitute a natural, unequally spaced order, the ordinal logistic regression model can be considered appropriate. Price-driven buyers were used as the reference category; given the small sample size, quality/taste and environmental/welfare drivers were each combined into a dummy variable. The model covered 84 respondents with complete data.

Table 9. Ordered logistic regression: determinants of WTP (MASS::polr, n = 84; reference = price driven buyers)

Variable	Coefficient	t value	Approx. p value / OR
Driver: quality or taste (vs price driven)	2.183	2.949	p ≈ 0.003; OR ≈ 8.9
Driver: eco / animal welfare (vs price driven)	1.996	2.133	p ≈ 0.03; OR ≈ 7.4
Driver: local product (vs price driven)	1.520	2.060	p ≈ 0.04; OR ≈ 4.6
Current milk drinker (vs non drinker)	2.303	2.116	p ≈ 0.03; OR ≈ 10.0
Alternative purchase channel	2.253	2.492	p ≈ 0.01; OR ≈ 9.5
Consumption volume (ordinal, 0–3)	−1.077	−2.892	p ≈ 0.01; OR ≈ 0.34

Reference category for purchase driver: price driven buyers. Results are exploratory given n = 84 and convenience sampling. The model significantly improves on the intercept-only *specification, but the proportional-odds assumption was not formally tested; full diagnostics, odds ratios and the excluded "other" category are reported in Appendix D.*

Quality/taste is the strongest purchase driver for higher WTP (OR ≈ 8.9), followed by eco/welfare (OR ≈ 7.4) and local product (OR ≈ 4.6). Milk drinkers and alternative-channel buyers show even stronger effects, while higher consumption is associated with lower WTP. The sample is small (n = 84), so these results should only be treated as indicative. Within this sample, the more relevant B2C group appears to be consumers who are not mainly driven by price, already buy through non-supermarket channels, and do not belong to the highest consumption group.

The survey gives limited support for a B2C price of €2.10/L. For this reason, the financial model also considers €1.90, €2.30 and €2.50/L in the sensitivity analysis (Table 14, Appendix E).A lower price may attract more customers, but there is little room for a large reduction because the milk purchase cost is already €1.65/L. Although 26.6% of respondents stated a WTP of at least €1.50/L, this level would not cover the milk cost itself. The survey also did not test €1.80 or €1.90 directly, so it cannot show how consumers would respond at these prices. B2B may fit the delivery model better because each HORECA customer can purchase a larger volume per stop. Household WTP is also less directly relevant to this channel. However, no Lugo HORECA survey was conducted, so actual willingness to buy at the proposed B2B price remains unknown.

4.5 Competitive Landscape: Milk and Dairy Substitutes in the Lugo Market

The competitive analysis considers two levels of choice. The first is between conventional and organic milk. The second is the choice between brands within the organic or premium fresh-milk category. Retail prices were checked in Lugo in June 2026, while the full comparison is reported in Appendix A.

At €2.10/L, the proposed price is more than twice the approximately €1.05/L charged for conventional Galician milk in Gadis, Eroski and Mercadona. The survey results show a similar price constraint: 68.1% of respondents stated a maximum WTP of €1.50/L, while only around 3% of those selecting “local product” as their main purchase factor reached €2.00/L. Local origin therefore appears to have value, but not enough to explain the full price difference for most respondents.

The price comparison is different within the organic and premium fresh-milk category. Arqueixal is priced at about €2.00/L and already delivers organic fresh milk in glass bottles to Lugo twice a week. Xanceda is available in supermarkets for approximately €1.89–€2.15/L, although it uses plastic bottles. Arqueixal is the more similar competitor in terms of product and delivery format: both are organic, glass-bottled and home-delivered. Xanceda, by contrast, sits closer to the proposed €2.10/L price range, though it is sold in plastic bottles through supermarkets rather than delivered directly.

Sen Máis creates a more direct problem for the proposed service because the same milk is already available in neighbourhood shops in Lugo. Omar reports a retail price of about €2.10/L, although this has not been confirmed with Sen Máis. If that figure is correct, As Leiteiras would not be offering customers a cheaper or different product; the reason to switch would have to come from the delivery service itself. Existing Sen Máis buyers would therefore be a useful group to test first. They already know the product and are apparently paying close to the proposed price, but it is still unknown whether home delivery would be attractive enough to change how they buy it. Delivery times, minimum orders and bottle returns would need to be tested directly with them. It is not yet known whether Sen Máis would agree to an exclusive or semi-exclusive home-delivery arrangement in Lugo. This would need to be discussed with the producer before any launch decision.

Oatly Barista (€2.00–€2.36/L retail) provides a reference for another premium product used in cafés. However, its retail price is not directly comparable with a HORECA procurement price, which was not available. Plant-based products are therefore not used for the main B2B comparison.

The comparison shows that As Leiteiras would not enter Lugo with a completely new offer. Arqueixal already delivers organic fresh milk in glass bottles, while Sen Máis milk is also available through local shops. The main difference would therefore have to come from the delivery service rather than from the milk itself. The survey points to a relatively narrow B2C group, particularly quality- and taste-oriented consumers and those already buying outside mainstream supermarkets. However, the survey does not provide enough evidence to estimate the actual number of customers in this segment.

4.6 Supplier-side Implications and Channel Fit

As Leiteiras would depend strongly on Sen Máis because it would distribute the milk rather than produce it. If the same milk continues to be sold in Lugo shops at around €2.10/L, the delivery service would have limited product level differentiation. Convenience and delivery reliability would then become more important. I also visited several Lugo stockists informally and found that Sen Máis was not always available. Some shops appeared to receive stock only once a week. This may give scheduled home delivery some practical value, although the observation was informal and should not be treated as a systematic retail survey.

The financial model does not assume any preferential arrangement with Sen Máis. It keeps existing shop sales in place and assumes that As Leiteiras receives no exclusivity. The unresolved commercial terms with Sen Máis are therefore not what drives the Base Case result. An exclusive home-delivery agreement could strengthen the commercial position, but the current no-go conclusion is not conditional on obtaining one. Before any launch decision, Omar would still need to confirm whether Sen Máis is willing to support a separate delivery channel and under what conditions.

4.7 B2B / HORECA Channel: Rationale, Delivery Economics, and Viability Scenarios

This section does not use the household consumer survey. It is based on Omar's project ideas, an interview with Ramón Barreiro, and several illustrative scenarios using the model's unit economics. Barreiro was a co-founder of a similar milk-delivery project in Santiago de Compostela, which operated from 2010 to 2016. No separate HORECA survey was carried out in Lugo. The results in this section should therefore be treated as a feasibility assessment rather than direct market evidence.

The driver is the single largest fixed cost, and it must be covered before any net income (see Section 4.8 for the full cost structure). B2B carries a lower margin per litre than B2C, because of the volume discount and absent retail intermediary, but delivers more litres per stop. Whether this trade-off is worthwhile depends on the size of the account: illustrative route-capacity comparisons (Table 11, Appendix E) show small B2B accounts barely matching the B2C route's daily contribution, while medium and large accounts clearly exceed it. B2C volume alone cannot sustain the model; B2B accounts must carry the majority of delivered litres.

Santiago's project bore little resemblance to the As Leiteiras model on the production side: the founders owned the cows, processed the milk themselves, and ran their own dispensing machines, covering steps that Sen Máis and Omar would split between them here. At its peak, the eight machines sold a combined 1,500–3,000 L/day, according to Barreiro. That figure says more about how large the Santiago operation grew than about what Lugo could sell, since the two businesses were built on different foundations. The case is more useful for understanding the operating burden. Barreiro explained that the founders were responsible for the whole process, from milk production and processing to sales and delivery. In his account, the workload was an important reason for leaving the business. This experience supports keeping the proposed As Leiteiras model focused on distribution rather than dairy production.

Barreiro reported that his five current restaurants in Santiago use Sen Máis milk. This provides some qualitative evidence that the product can be used in a HORECA setting, although it does not show how large the potential market would be in Lugo. This is interview evidence from one business owner and should not be generalised to the Lugo HORECA market. It does, however, show that Sen Máis milk is already being used by restaurants in another Galician city. The Santiago project also had a feature that As Leiteiras would not have: the dispensing machines themselves were visible to consumers. They may have helped the project attract attention and word of mouth. Without these physical sales points in Lugo, As Leiteiras would probably need to approach potential B2B customers more directly. Further detail on the Santiago case and Barreiro's account appears in Appendix E.

Table 12. Financial model-aligned channel assumptions at stable operation

Channel	Stable units	L/unit/week	Annual litres	Volume share
B2C households	180	3	28,080	42%
B2B / HORECA accounts	30	25	39,000	58%
Total	**—**	**—**	**67,080**	**100%**

B2C: 180 active households × 3 L/week × 52 weeks = 28,080 L/year. B2B/HORECA: 30 accounts × 25 L/week × 52 weeks = 39,000 L/year. The 25 L figure refers to total weekly volume per active HORECA account, not litres per delivery occasion. Volume shares are rounded. These figures align with the financial model's stable-year planning layer.

B2B accounts are few in number but provide most of the expected sales volume: 30 accounts generate 58% of annual litres under the stable-year assumptions. This means that B2B is not a secondary channel in the model. The route depends heavily on obtaining enough HORECA volume.

The sensitivity analysis uses the same stable-year assumptions. For B2C, it refers to the demand proxy calculated in Section 4.4. For B2B, it uses Omar's reported pool of 272 possible establishments. This figure includes hospitality and some specialty-retail businesses, so it should not be treated as a verified HORECA market size. The scenarios test different numbers of customers reached from these two groups.

Table 13. Financial-model-aligned sensitivity analysis: stable-year scenarios

Scenario	B2C customers	B2B accounts	B2B conv. (272 pool)	Annual litres	Operating costs/year	Revenue/year (net of VAT)	EBITDA/year
Conservative	120	15	5.5%	38,220	€110,623	€72,488	−€38,135
Base case	180	30	11.0%	67,080	€161,359	€129,825	−€31,534
B2B upside	180	45	16.5%	86,580	€195,487	€166,388	−€29,099
B2C upside	240	30	11.0%	76,440	€177,835	€148,725	−€29,110
Combined upside	240	45	16.5%	95,940	€211,963	€185,288	−€26,675

B2C: active customers × 3 L/week × 52 weeks. B2B: active accounts × 25 L/week × 52 weeks. B2C price €2.10/L. The Base and upside scenarios use the Base B2B price of €1.95/L, while the Conservative scenario follows the financial model's conservative B2B price of €1.85/L. Purchase cost is €1.65/L. Operating costs and EBITDA reflect the relevant scenario cost structure from the financial model. Revenue, operating costs and EBITDA figures are annual. A B2C conversion rate is not shown because the two candidate denominators are not comparable: the 637-unit population-based proxy (Section 4.4) is an individual-level stated-

WTP figure, while active B2C customers here are household accounts; dividing one by the other would not be a valid conversion rate. B2B conversion is shown because both active accounts and the 272-account B2B prospect pool are counted in the same units.

Both upside scenarios remain EBITDA-negative; B2B increases litres delivered per stop but does not solve profitability alone. Varying the B2C price instead, holding volumes constant (Table 14, Appendix E), narrows but does not close the gap, since most operating costs are volume- rather than price-driven.

HORECA and household clients differ in ways that matter for go-to-market: The financial model assumes 25 L/week per B2B account and 3 L/week per B2C household. The B2B assumption is therefore about eight times the B2C household volume. B2B deliveries are assumed to be more regular than household deliveries, but this has not been tested with Lugo businesses. The model also assumes that B2B customers pay within 30 days. At stable operation, this creates an estimated receivables balance that is not yet included in the cash flow forecast. The B2B results should therefore be read with these operating and payment assumptions in mind.

Supply may create a separate constraint. The current working estimate is that around 500 L/day would be available from Sen Máis, although this figure still needs to be confirmed. The financial model requires about 965 L/day to reach the lower-bound full-cost EBITDA break-even point. If the 500 L/day estimate is correct, the business would reach the supplier limit well before reaching that break-even volume.

These assumptions make the B2B case less certain, but they are not what produces the negative result. Table 13 remains EBITDA-negative in every scenario tested, including cases with more B2C customers or more B2B accounts. The more basic problem is that the sales volumes tested are still too low for the current cost structure. On that evidence, I would not recommend committing to a dedicated refrigerated van and a full-time driver under the present launch model. A smaller pilot or a different operating model would need to be analysed separately. The current results should not be interpreted as evidence that such an alternative would be profitable.

4.8 Link with the Financial Model

Table 13 uses the same main assumptions as the financial model, including prices, consumption rates and stable-year operating costs. Its EBITDA results can therefore be compared with the stable-year figures in the workbook. Selling prices are assumed to include VAT, although this has not been confirmed with Sen Máis, and the 4% milk VAT is removed when calculating net revenue. Detailed inputs remain in the financial model and are not repeated here.

Table 13 combines the B2C demand proxy with the B2B prospect figures described above. The financial model itself uses 180 B2C households and 30 B2B accounts in the stable year. Route A is the route used in the financial model, while Routes B and C are discussed only as possible alternatives and are not calculated in the workbook. The daily trip to Sen Máis is also treated separately from customer deliveries.

There is no primary Lugo data on HORECA price acceptance or customer conversion. For this reason, the analysis tests several scenarios rather than estimating one expected level of B2B demand. Even the scenarios with 45 B2B accounts remain EBITDA-negative, so the lack of precise demand data does not remove the main financial problem identified by the model.

5. Discussion: Reachable Market Segment, Positioning, and Launch Implications

The marketing and financial results need to be separated here. The marketing analysis identifies a relatively narrow group of consumers who may be more interested in the product. However, targeting this group more precisely does not solve the financial problem in the current model. Even with more favourable customer scenarios, the tested cost structure remains unprofitable. The difficulty comes from several factors occurring at the same time. The proposed price is around twice the price of mainstream supermarket milk, home delivery is almost unused among the survey respondents, and only a small share stated a WTP close to the proposed price.

The survey also shows an interesting difference between the segments. The local and eco group is the largest, but its WTP is lower than that of the quality and taste group. That difference may partly reflect the Lugo market. Brands such as Larsa and Feiraco already give consumers a familiar local option at much lower prices, so local origin by itself seems unlikely to justify the proposed premium. If B2C testing goes ahead, the more promising respondents are those who prioritise quality or taste and those who already buy outside supermarkets. The regression points in the same direction, but with only 84 complete observations it is better treated as supporting evidence than as a precise target profile.

Local origin alone therefore cannot justify a €2.10/L price: it is the factor the largest group cites, yet almost none of that group accepts a price near this level. A positioning built only on "local" or "organic" would not fit what the data show. Freshness, taste and returnable glass packaging provide additional elements that could be tested with customers. Local origin and organic certification would still be relevant, but the survey does not show that either is sufficient on its own to support the proposed premium. One possible positioning for further testing is therefore a premium fresh milk service built around home delivery and returnable glass packaging. Delivery is important because customers can already obtain Sen Máis milk through local shops. The proposed service must therefore offer some additional convenience if customers are expected to change channel. The commercial arrangement with Sen Máis would also need to be clarified before launch.

B2B cases are based on a different logic and should be evaluated separately: the sales of a coffee shop or restaurant are equivalent to several B2C households, so the route economics depends more on B2B density than on the number of households. The Santiago case and Barreiro's continued use of Sen Máis suggest that HORECA demand for this type of product can exist, but no HORECA survey was carried out in Lugo itself (Section 4.7), so this remains an inference rather than confirmed local demand. The open question before launch is therefore twofold: whether that demand actually holds in Lugo, and whether enough accounts can be secured within a concentrated delivery area to make the route viable before the B2C base grows large enough to support it.

These patterns connect to three established strands in the literature reviewed in Section 2. The gap between stated and observed price acceptance is consistent with the hypothetical bias documented in the stated preference literature (Murphy et al., 2005), while Moser et al. (2011) review evidence that credence attributes can be associated with higher willingness to pay. The Lugo data show why it arises here: credence attributes such as organic or local origin only command a premium when consumers cannot already satisfy the relevant value through a cheaper available substitute, which Larsa and Feiraco already provide. This has a direct positioning implication, since retail branding shapes customer perceptions and store choice (Ailawadi & Keller, 2004). An advantage built on being organic, local and bottled in glass is easy to copy, whereas one built on the customer benefit of receiving very fresh local milk conveniently depends on reliable delivery, route density and a smooth ordering experience. Any sustainable advantage is therefore more likely to lie in operational than in product differentiation, since the physical product is identical to what Sen Máis already sells through retail channels. From a short food supply chain perspective (Kneafsey et al., 2013), As Leiteiras mainly adds value by making the product easier to access rather than by changing the product itself. The premium price alone is not enough to make the model work if delivery stops are widely dispersed. Route concentration is also important for controlling delivery costs. The Santiago case provides some support for the importance of HORECA customers, although Barreiro did not provide an exact split between B2C and B2B sales.

Lugo is also a relatively small market for a premium product. The analysis identifies only a limited group of consumers who appear willing to consider the proposed price. For this reason, the results from Lugo are not enough to show that the same model could be transferred easily to other Galician cities.

6. Conclusions and Recommendations

The financial model does not support a full-scale launch under the current assumptions. In particular, the model includes dedicated refrigerated vehicles, full-time drivers and daily collection from Sen Máis, and the expected sales volume is not enough to cover this cost structure. The current result applies to the full-cost model tested here, not to every possible version of As Leiteiras. Over the five-year window, the discounted project NPV is approximately −€171,869 (EBITDA-based proxy, excluding financing), reinforcing the EBITDA-based no-go conclusion. Before abandoning the idea, two relatively simple checks would still be useful. Interviews with cafés and restaurants in Lugo could show whether there is real B2B interest, and a direct discussion with Sen Máis could clarify the available supply and commercial terms. Neither step would make the business viable by itself, but both would replace important assumptions with better evidence.

The main difficulty remains the same. The expected sales volume is too low for the current cost structure. Household demand at around €2.10/L is limited in the survey, while B2B accounts provide more litres per customer but still do not bring the tested scenarios into positive EBITDA. For that reason, I would not move to a full-scale launch at this stage. B2B demand should be checked directly first, and any lower-cost operating model would need to be analysed separately.

Table 15. Research questions and answers, in brief

Research question	Answer
SQ1 (Costs and break-even): how many bottles per day, at what price, to cover costs?	Roughly 965 L/day is needed to break even at €2.10/L B2C and €1.95/L B2B, about 3.6 times the modelled stable-year volume. At the model's average of 5.375 litres per stop, this equates to approximately 180 stops per day, well beyond the 68-stop limit for a single driver. Reaching it would also require a second driver and higher route and fuel costs, which the 965 L/day figure does not include, so the true break-even point is higher still. None of the tested scenarios reach this volume. Raising the price does not resolve this either: even at €2.50/L, the model still shows a stable-year loss of approximately €21,490.
SQ2 (Hiring): full-time or part-time driver from the start?	Neither option closes the gap on its own. A part-time driver (0.5 FTE) does improve on the Base Case, reducing the loss to approximately €24,487 per year, but the result remains a loss.
SQ3 (Who to sell to): B2B or B2C, which is more efficient?	B2B matters more than B2C: just 14% of accounts generate 58% of stable-year volume. It is not sufficient on its own, however; even the B2B-upside scenario remains EBITDA-negative.
SQ4 (Sensitivity): to price, litres/stop, and fuel cost?	EBITDA remains negative across every scenario that was directly recalculated: four price points from €1.90 to €2.50/L, and the five demand/channel scenarios in Table 13. Litres-per-stop and fuel cost were tested only for their operational and cost effects, not as full EBITDA recalculations, but both point in the same direction: fewer litres per stop increases the required stops beyond single-driver capacity, and higher fuel costs would only widen the loss. Within what was directly tested, the no-go conclusion holds.
SQ5 (Scaling up): could the model scale to other Galician cities?	The study does not establish whether the model can be replicated in other Galician cities. Replication would require, at minimum, a suitable supplier, sufficient B2B density and acceptable B2B pricing.

For SQ5, Table 15 identifies three minimum conditions for replication. In Lugo, B2B density is one of the main constraints identified by the analysis. This study does not test whether the same constraint would apply in other Galician cities.

If the project is reconsidered, Lugo city should remain the first area to test. It contains most of the Core and Tier 1 population and has a much higher population density than the surrounding municipalities. The additional B2C demand estimated from the Tier 1 population is relatively small. Expansion outside Lugo would therefore make more sense only if enough B2B customers could be added to the same route or if the additional stops could be served efficiently.

At this stage, two pieces of primary information would be more useful than further desk research. First, a small number of structured interviews with café and restaurant owners in Lugo could test whether there is real interest in the B2B offer. Second, Sen Máis should be contacted directly to clarify the possible supply arrangement and whether existing retail sales would continue. I would obtain this information before considering a van purchase or hiring a driver. These interviews could improve the assumptions used in the model, but they are unlikely to reverse the present result by themselves. Even the more favourable demand scenarios tested in Table 13 remain EBITDA negative.

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Appendix A: Competitive Landscape — Extended Analysis

This appendix expands the competitor matrix (Table 10) and brand comparison summarised in Section 4.5, moved here to keep the main results section shorter. Table numbering follows the main text.

A.1 Analytical framework

For Lugo market, I separate the competitive comparison into two questions. The first is whether consumers would move away from conventional milk towards an organic or premium option at all. The second is which product they might choose once they are already considering that smaller category. This distinction matters because As Leiteiras competes both with low-priced supermarket milk and with closer alternatives such as Arqueixal and Xanceda. Sales channel and plant-based products are included as additional points of comparison.

The evidence used here comes from several different sources. Retail prices were checked in Lugo in June 2026 at Gadis, Eroski and Alcampo, while Arqueixal's delivery route was taken from the company's website. The Sen Máis retail price is different because it was reported by Omar and has not been independently verified. Brand mentions come from the 94-response Lugo survey conducted in May–June 2025.

A.2 Level 1: Conventional vs. organic, the category switching decision

For most consumers, the first price comparison is with conventional milk: Larsa, Feiraco and LEYMA sell for around €1.05/L in supermarkets, and Hacendado runs under €1.00/L. At €2.10/L, As Leiteiras would cost more than twice that. The question is whether enough respondents in the survey are actually willing to pay that gap.

The survey shows substantial price sensitivity. Around 68.1% of the full sample stated a WTP no higher than €1.50/L, while only a small share reached €2.00/L. Respondents selecting local origin as an important purchase factor also showed limited acceptance of the higher price. Local origin therefore appears to have some value for respondents, but it does not by itself explain the price difference between As Leiteiras and conventional Galician milk.

The price problem is also linked to the strong presence of established Galician brands. Some of the proposed attributes are already available through competitors. For example, Xanceda offers fresh organic milk through supermarkets, while Arqueixal offers a format closer to the proposed As Leiteiras service. The current evidence does not show whether consumers would pay more for the particular combination offered by As Leiteiras, so this would need to be tested directly.

A.3 Level 2: Choice among organic fresh milk options

A.3.1 The competitive set

Table 10 compares the identified organic and premium milk alternatives using five characteristics: freshness, packaging, organic certification, availability in Lugo and sales channel. A ✓ indicates that the attribute is present and a ✗ that it is absent.

Table 10. Competitor matrix: Lugo milk and dairy substitute market (June 2026)

Brand / product	**Price (€/L)**	**Fresh**	**Glass bottle**	**Organic certified**	**Available in Lugo**	**Strategic note**
Hacendado (Mercadona)	€0.82–0.96	✗	✗	✗	Yes: Mercadona stores	Price floor; 19 survey mentions; dominant mass market reference

Brand / product	Price (€/L)	Fresh	Glass bottle	Organic certified	Available in Lugo	Strategic note
Larsa / Feiraco / LEYMA	€1.05	✗	✗	✗	Yes: Gadis, Eroski, Mercadona	Top survey brands by mention count: Larsa 23 (24.5%), Feiraco 16 (17.0%), LEYMA 11 (11.7%). Multiple selections permitted; Galician identity but supermarket UHT
Carrefour Bio	€1.26	✗	✗	✓	Yes: Carrefour	Entry level organic; mass channel; no local provenance claim; UHT
LEYMA Natura / Larsa fresca	€1.37	✓	✗	✗	Yes: Gadis, Eroski	Galician fresh, plastic bottle; not organic; benchmark for fresh non organic
Xanceda (fresh semi skimmed)	€1.89–2.15	✓	✗	✓	Yes: Gadis (€1.89), Eroski/Alcampo (€2.15)	Closest organic fresh rival; plastic PET; 2 survey mentions (1 at WTP ≥€2.00, 1 at €1.50–2.00)
Arqueixal	€2.00	✓	✓	✓	Yes: direct route Tue/Fri via Guntín	Most direct format competitor: Galician, organic, glass, fresh, home-delivery to Lugo

Brand / product	**Price (€/L)**	**Fresh**	**Glass bottle**	**Organic certified**	**Available in Lugo**	**Strategic note**
Sen Máis (retail)	€2.10	✓	✓	✓	Yes: tenda de barrio Lugo	As Leiteiras' own supplier; same product at same price; channel conflict risk
As Leiteiras (proposed)	€2.10	✓	✓	✓	Proposed: home-delivery	Potential differentiation through a more concentrated city-based service, easier ordering and greater delivery flexibility; same product/price as above
Hacendado bebida avena	€0.80	✗	✗	✗	Yes: Mercadona	Main plant-based substitute; own label oat drink; €4.80/6 pack
Oatly Barista	€2.00–2.36 (retail)	✗	✗	✗	Yes: Carrefour, Eroski, HORECA	Premium oat milk; HORECA substitute; exact procurement price unconfirmed

Sources: Gadis online (gadisline.com), Eroski supermercado online, Alcampo online, Arqueixal website (arqueixal.es), Sen Máis retail price per Omar (working assumption, unconfirmed at source), and Oatly Barista retail listings on Amazon Spain and Hipercor. ✓/✗ cells shaded green/red for readability. Price ranges reflect inter retailer variation where observed.

A.3.2 Arqueixal: the most direct format competitor

Arqueixal is the closest direct competitor because it combines all five premium attributes: fresh milk, returnable glass bottles, organic certification, Galician origin and delivery to Lugo. Its price of €2.00/L is slightly below the proposed As Leiteiras price of €2.10/L. Arqueixal already delivers on Tuesdays and Fridays along the Palas de Rei, Guntín and Lugo route, so it is already offering a very similar product and delivery format in the area.

Notably, Arqueixal received zero mentions in the 94 respondent survey. Given the small, non-random survey sample, this cannot be taken as evidence of low awareness in the broader Lugo population.

The two models differ mainly in how the delivery service is organised. Arqueixal operates as a farm to consumer direct model with delivery on fixed days twice a week. As Leiteiras proposes a more city-focused service than Arqueixal's fixed Tuesday/Friday route, but the analysis does not establish whether easier ordering or greater scheduling flexibility would be enough on their own to attract customers away from Arqueixal. Arqueixal also combines rural tourism and cheese production with its milk business, giving it revenue diversification that As Leiteiras does not have.

A.3.3 Xanceda: the organic fresh benchmark

Xanceda (€1.89–2.15/L; Table 10) is the closest supermarket-channel organic rival, but in a plastic PET bottle rather than glass. Two respondents mentioned Xanceda: one reported WTP of at least €2.00/L and the other €1.50–2.00/L. With only two observations, no broader consumer pattern can be inferred. As Leiteiras's €2.10/L falls within this range, close to its upper bound, so the glass bottle and home-delivery service are what would need to justify the price to a consumer, not a price premium over Xanceda; the two would in any case never appear on the same shelf.

A.3.4 Sen Máis retail: the channel conflict

An unusual feature of the competitive landscape is that As Leiteiras's direct supplier, Sen Máis, is also a competitor in the Lugo retail market. Sen Máis is said to sell its glass-bottled fresh organic milk through neighbourhood shops (tendas de barrio) in Lugo at €2.10/L, a figure Omar reported, not independently verified with the producer, and treated here as a working assumption. This is the same price As Leiteiras proposes to charge for home-delivery of the same product.

This creates a potential channel conflict: a consumer who discovers As Leiteiras's home-delivery service might equally discover that the same product is already available at their local shop. Existing Sen Máis buyers could be useful for further customer research because they already know the product and, if Omar's reported price is correct, already pay close to €2.10/L. However, the current evidence does not show whether they would switch from neighbourhood shops to home delivery. Delivery arrangements and bottle returns would also need to be tested.

The commercial arrangement with Sen Máis is still unclear. It is not known whether existing shop distribution would continue unchanged or whether any form of home-delivery exclusivity would be possible. This should be clarified before an investment decision.

A.4 Plant-based Substitutes

Only two survey respondents mentioned plant-based milk, so it plays a minor role in the B2C analysis. Oatly Barista (€2.00–€2.36/L retail) provides some price context for premium café products, but no verified HORECA procurement price is available. Plant-based products are therefore not included in the main B2B comparison.

A.5 As Leiteiras' competitive position

Arqueixal already delivers organic milk in glass bottles to the city on fixed days, so As Leiteiras would not be entering Lugo with a genuinely new offer. Product differentiation is limited for the same reason: the milk itself is Sen Máis's, already sold through some Lugo shops, and Arqueixal already provides a comparable delivery format. Any real difference would therefore have to come from how the delivery service is organised, not from the product. The reachable B2C group therefore appears narrow, concentrated mainly among quality- and taste-driven respondents and those who already shop outside mainstream supermarkets. Existing premium-milk buyers are a further candidate for future research, though the survey cannot size how many such customers exist across Lugo.

A.6 Remaining Competitive Questions and Resolved Price Check

Several questions remain unresolved and are not modelled further here. Arqueixal's actual retail presence in Lugo tendas de barrio is unconfirmed, and whether Sen Máis would grant exclusive home-delivery rights in Lugo or continue supplying neighbourhood shops in parallel is unresolved and requires a direct conversation with the producer. Xanceda's whole-milk (entera) price, more relevant to As Leiteiras's product than the semi-skimmed price collected, has also not been confirmed. At €1.31/L, Alcampo's Puleva Eco organic milk is the lowest-priced organic option identified in the sample, while As Leiteiras and Arqueixal are positioned at the higher end of the market, with prices ranging from €2.00 to €2.10/L.

Appendix B: Market Geography — Extended Analysis

This appendix contains a route map (Figure 1), route design plans (Table 2), and a detailed discussion of the limitations behind the shorter treatment in Section 4.2. To make the main text more concise, the relevant content has been moved here. The original numbering of tables and figures is retained.

B.1 Defining the geographic market

The 94-response consumer survey was not designed to represent different geographic areas. It was collected through convenience or snowball sampling rather than random sampling. Most respondents came from Lugo city: 85 people (90.4%), compared with 2 from O Corgo, 2 from Guntín, and 5 who did not report their location. The survey should therefore not be treated as representative of either Lugo city or the surrounding municipalities. For this reason, this section uses public population and infrastructure data to assess the area that could realistically be served.

Route feasibility depends mainly on logistics, particularly distance, road access and daily driving time, and on customer density, since more concentrated customers allow more stops within a shorter route. The location of Sen Máis is also important. The farm is in A Cernada, near Palas de Rei, around 34 km from Lugo by road. This pickup should therefore be treated as a separate supply journey rather than as part of the Lugo delivery route.

B.2 Municipality level assessment

The Comarca de Lugo comprises eight municipalities. The metropolitan transport agreement signed by the Concello de Lugo and the Xunta de Galicia extends functional links to Baleira and Castro de Rei as well. Table 1 assesses the eight comarca municipalities on three criteria:

population size (to estimate potential customer volume), road distance from Lugo city (to estimate daily driving time and fuel cost per route), and population density (to estimate how many stops per route kilometre are achievable). Each municipality is then assigned a delivery tier reflecting whether these factors, taken together, support viable route economics at launch or only at larger scale. Figure 1 shows the route from the Sen Máis farm at A Cernada to Lugo city centre (fastest route via LU 232: 34.3 km, approximately 39 minutes by car).

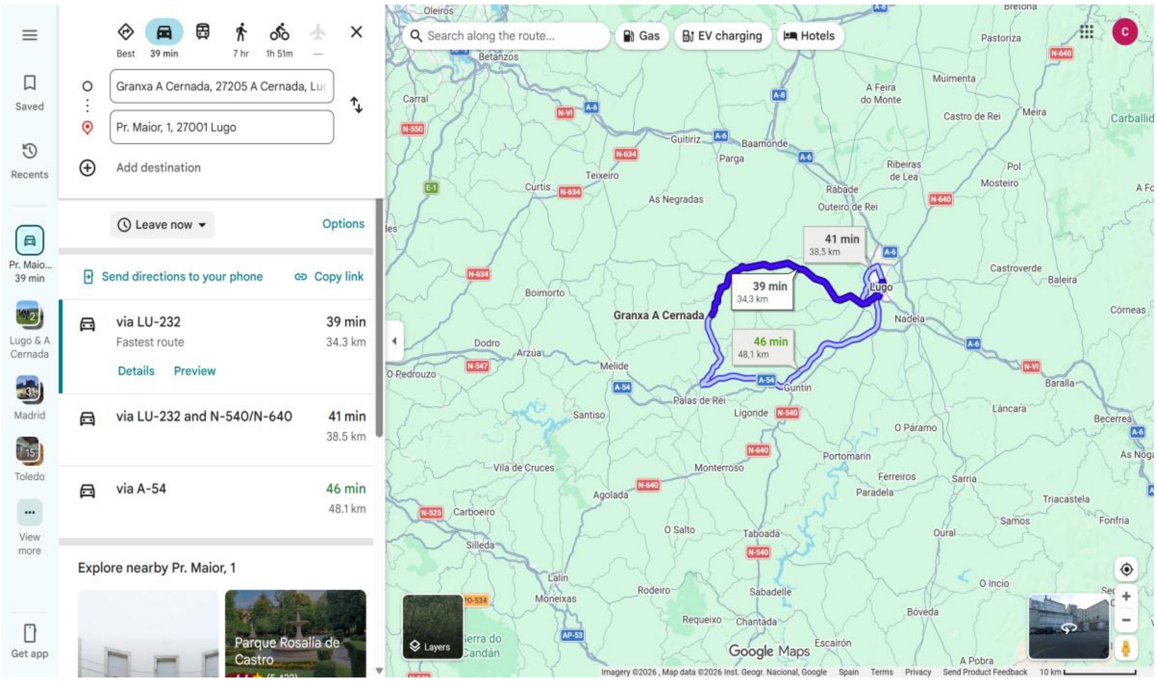

Figure 1. Route from Granxa A Cernada (Sen Máis farm) to Lugo city centre (Praza Maior). Fastest route via LU 232: 34.3 km, approximately 39 minutes by car. Source: Google Maps, June 2026.

Figure 1 shows the Google Maps route from A Cernada to Lugo city centre at approximately 34.3 km. The financial model uses a more conservative 46 km one way routing assumption for fuel and driver time modelling.

Lugo city accounts for about 88.6% of the combined Core and Tier 1 population of 112,238. Expanding into any one suburban municipality would therefore add only a relatively small number of potential customers. Outside Lugo, the route becomes less attractive quite quickly. Lugo has 302 inhabitants per km², compared with 40 in Outeiro de Rei and only 12 in Friol. This does not show the actual distance between customers, but it suggests that stops outside the city are likely to be more dispersed.

O Corgo and Guntín illustrate the problem. Both are around 18–19 km from Lugo, and neither is on the Sen Máis pickup route. Adding either municipality would therefore mean extending the delivery route in a separate direction rather than combining it with the milk collection trip. Unless future demand is concentrated enough to justify the extra distance, particularly through B2B accounts, there is little operational reason to include them.

Rábade is different from the other smaller municipalities. It has only 1,499 residents, but they are concentrated in an area of about 5.2 km², giving a population density of roughly 290 inhabitants per km². This is close to Lugo's 302 and much higher than most of the surrounding areas. From a routing perspective, this is one reason to keep Rábade within Tier 1 rather than treating it like the more dispersed municipalities. Population density alone cannot show whether actual customers would be clustered closely enough for efficient delivery, so this remains only an initial geographic advantage.

B.3 The Sen Máis pickup route and route direction

The model assumes that milk is collected each day from Sen Máis in A Cernada, about 34 km from Lugo. This journey is required regardless of where customers are located, so the financial model records it separately from the distance travelled for customer deliveries.

O Corgo and Guntín are about 18 km and 19 km from Lugo, but neither is on the way to Sen Máis. Deliveries to these municipalities could therefore not be combined with the collection trip and would add extra distance to the customer route.

O Corgo and Guntín sit close enough to Lugo to be worth testing later, but only if customer concentration and B2B demand turn out to support it. Outeiro de Rei and Rábade to the north offer higher population density and direct A 6 motorway access, and neither direction carries a structural cost advantage over the other derived from the supply run.

B.4 Route design scenarios

Table 2 presents four route configurations, from the minimum viable urban core to a full comarca operation, with estimated daily driving distances and the conditions each requires to be financially justified.

Table 2. Route design scenarios and viability conditions

Route	Municipalities covered	Population reachable	Est. daily km driven	Condition for viability
Route A (Urban core)	Lugo city only	99,482	~40 km	Baseline scenario; highest available B2C density, but still requires B2B support for viability
Route B (Northern suburban)	Lugo + Outeiro de Rei + Rábade	106,339	~60 km	B2B accounts in Outeiro industrial zone justify extra km
Route C (Southern rural)	Lugo + O Corgo + Guntín	105,381	~70 km	Requires B2B anchor accounts; no pickup route synergy assumed
Route D (Full comarca)	All Tier 1 municipalities	112,238	~100 km	Requires multi driver capacity, high B2B density across municipalities, and established brand presence; route economics do not hold under single driver, single vehicle constraints

Daily km estimates are approximate, based on inter municipality road distances plus 40 km for internal Lugo city delivery routing, consistent with Route A and the financial model's city delivery km/day figure. Actual figures depend on customer clustering and stop density and would require route optimisation before launch.

Route A (Lugo city only) is the only configuration that does not require additional suburban B2B justification beyond the core Lugo route. However, even Route A still requires B2B support for overall financial viability. Routes B and C each add a suburban cluster and therefore require B2B anchor clients, sufficient customer clustering, or route density benefits to offset the additional cost. Route D is included for completeness but is not consistent with a lean Year 1 launch given the current single driver, single vehicle constraints.

The financial model quantifies Route A only, as the base case; Routes B, C and D are qualitative route-design scenarios discussed here for planning context and are not separately built into the financial workbook. Route C, covering O Corgo and Guntín, is included only as an illustrative scenario and is not modelled in the financial workbook. It should not be assumed to benefit from the Sen Máis pickup route. If this route were considered in the future, it would need enough B2B anchor accounts to justify the additional distance. Route B is discussed as a possible medium term expansion, conditional on signing B2B accounts in the Outeiro de Rei industrial zone, but it is likewise not quantified in the financial model.

B.5 Population-based Market Size at Municipality Level

This section defines the geographic population base only; the WTP-based demand proxy that applies to it is derived in Section 4.4, after the consumer profile and WTP analysis. The two relevant population figures are the Lugo city (Core) population of 99,482 and the Core plus Tier 1 population of 112,238.

Section 4.4 shows that adding the Tier 1 population increases the B2C demand proxy only slightly. On this basis, suburban expansion cannot be justified by additional B2C demand alone. B2B demand would need to provide an additional reason for extending the route.

B.6 Limitations

- Raw data for suburban towns is lacking. Most survey respondents live in Lugo city, and the convenience sample is not representative even of the city population. The data therefore cannot be used to estimate WTP or consumption patterns in O Corgo, Guntín or the other surrounding municipalities.
- Population density vs. stop clustering. Population size provides an indication of the potential number of delivery stops, but it does not capture how geographically concentrated these stops are. A higher concentration of households within a relatively

small area reduces travel time between stops and allows drivers to deliver more litres per hour. In contrast, rural municipalities may have a sufficient population but still have households spread across multiple small aldeas, making delivery routes less efficient.

- Route optimisation not modelled. The km estimates in Table 2 use inter municipality distances. Actual routes within Lugo city and along suburban corridors would require stop level optimisation. The financial model uses a flat per route distance assumption.
- Castro de Rei excluded. Castro de Rei is included in the metropolitan transport agreement with a population of approximately 4,800–5,000, but belongs to the Terra Cha comarca (not Comarca de Lugo) and lies roughly 30 km from the city. It is omitted from the matrix pending further geographic scoping.

Launch-market conclusion. Taken together, the geographic evidence identifies Lugo city as the only defensible launch area on route-feasibility grounds: this is the Core area, with the highest population density, the shortest routes, and no reliance on unconfirmed Tier 1 B2B anchor accounts. This is a geographic finding about where a launch would be feasible, not a recommendation to launch. Section 6 concludes that the full-cost launch structure is not currently financially viable regardless of geography. Tier 1 and Tier 2 municipalities are possible future expansion areas and should only be considered once sufficient B2B demand has been confirmed in Lugo city.

Appendix C: Consumer Profile — Extended Analysis

This appendix provides the supporting material for Section 4.3, including the survey coverage table (Table 3), the behavioural profiles of the two primary target segments (Table 5), the analysis of age, household structure and income, and the discussion of study limitations. This appendix gives the fuller version of these materials, so the main text can stay within the page limit for this thesis.

The profile in this section relies mainly on the 94-response Lugo survey. It can describe reported purchasing behaviour and purchase priorities, but not respondents' age, income or household composition because these questions were not included in the questionnaire. INE statistics are used only to describe the wider demographic context of Lugo and are not used to infer characteristics of individual survey respondents.

C.1 Survey Scope

For this analysis, consumer information is organised into three areas: sociodemographic characteristics, psychographic factors such as purchase priorities, and current buying behaviour. The Lugo survey provides useful information on the latter two, including what respondents value, what they buy, where they buy it and how much they consume. It provides very limited sociodemographic information because age, income and household composition were not collected. Table 3 shows which variables are available for each area.

Table 3. Survey coverage by consumer profile dimension

Dimension	Survey coverage	Key finding
Milk consumption	Direct (consume_leite)	86% currently drink cow's milk; 8 respondents (9%) do not but still state WTP
Weekly volume	Direct (cantidade)	48% consume 1–3 L/wk; 29% under 1 L; 14% 3–5 L; 3% over 5 L/day*

Dimension	Survey coverage	Key finding
Purchase channel	Direct (modo_compra)	88% buy at supermarket; 6% at neighbourhood shop; 1% home-delivery
Brand repertoire	Direct (marca, open multi select)	Larsa 23 (24.5%), Feiraco 16 (17.0%), LEYMA 11 (11.7%). Multiple responses allowed; national brands also mentioned.
Purchase driver	Direct (motivos, open text)	Local product 32%; quality 21%; price 20%; taste 12%; animal welfare 6%
WTP	Direct (closed brackets)	See WTP section; 3.2% state WTP ≥ €2.00/L
Sociodemographic (age, gender, income, household)	NOT collected	Major gap; relevant for income capacity, household level volume, delivery adoption, and route economics; addressed only through external population data and inference.

Source: Survey instrument design assessment. The survey was conducted through a Google Forms questionnaire distributed by the project founder. The "over 5 L/day" category uses the original wording of the questionnaire, which specified a daily rather than weekly unit. This appears to be a wording inconsistency in the survey. Since only three respondents (n = 3) fall into this category, these results should be treated with caution and are considered separately in the WTP analysis.

The survey provides a relatively good understanding of consumers' behaviour and psychographic characteristics. Its main limitation is the lack of sociodemographic information, particularly on age, household composition, and income. As a result, these characteristics cannot be assessed directly from the survey. Where they are relevant to the analysis, INE population data are used to provide additional context, rather than to make precise claims about the target consumer.

C.2 Behavioural profile

C.2.1 Current milk consumption

Of the 94 respondents, 81 reported drinking cow's milk, 8 did not, and 5 did not answer the question. The most common consumption range was 1–3 litres per week (47.9%), followed by less than 1 litre (28.7%) and 3–5 litres (13.8%). Three respondents selected the "more than 5 litres per day" option.

These figures refer to individual respondents rather than households, since household size was not collected. They should therefore not be converted directly into household demand. The final response option also uses a daily rather than weekly unit, unlike the other categories, so the three responses in that category are treated cautiously.

C.2.2 Purchase channel

Milk purchasing was heavily concentrated in supermarkets: 88% of respondents used them, compared with 6% who bought from neighbourhood shops. Only one respondent reported using home delivery. The six respondents using neighbourhood shops already buy outside the dominant supermarket channel, which is a relevant detail. However, the survey does not show whether they would be more willing to switch to home delivery.

C.2.3 Brand repertoire

Respondents could name more than one milk brand. Among the Galician brands, Larsa was mentioned most often, by 23 respondents (24.5%), followed by Feiraco (16), LEYMA (11), Deleite (10) and Leche Río (7). These five brands received 67 mentions in total.

These percentages are not market shares. Because respondents could select several brands, they only show how often each brand was mentioned in the sample. National and retailer brands were also common, with 51 mentions in total: Hacendado/Mercadona (19), Central Lechera Asturiana (11), Pascual (10), Carrefour (6) and Eroski (5). Sen Máis was mentioned once, together with Quintián and Xanceda in the same reply.

Galician brands were frequently mentioned, while "local product" was also the most common purchase factor in the survey (32%). These two results are consistent with a strong familiarity with regional dairy products in the sample. However, neither result shows that respondents would change from their current purchasing channel to home delivery.

C.3 Psychographic profile: three consumer segments

The open-text question on purchase drivers was answered by 91 respondents. The responses were grouped manually according to the main factor mentioned by each respondent. This produced three broad consumer groups. This is a descriptive classification rather than a statistical segmentation. Table 4 summarises the three groups.

C.3.1 Quality and taste seekers (n = 31, 33%)

For this group, quality or taste was the main reason given for buying milk. It also had the highest WTP of the three groups: 42% were willing to pay at least €1.50/L and 6% selected €2.00/L or more. The brands mentioned include both Galician names such as Larsa, Feiraco and Deleite, and national brands such as Pascual and Central Lechera Asturiana. This suggests that their choices are not driven by local origin alone.

This group is the most relevant starting point for B2C testing because its stated priorities are closer to the proposed product positioning. Taste and product quality can be tested directly, while freshness, local origin and returnable glass bottles could be included as additional features of the offer. The current survey cannot show how much these individual attributes affect WTP because they were not tested separately. Price remains the main limitation: only 6% of this group selected €2.00/L or more, so support for a price close to €2.10/L is still weak.

C.3.2 Local and eco advocates (n = 38, 40%)

This is the largest of the three groups. Respondents in it mainly referred to local origin, organic production or animal welfare when explaining their milk purchases. Their WTP is lower than that of the quality and taste group: 21% selected at least €1.50/L and only 3% reached €2.00/L. These preferences are relevant to the As Leiteiras concept, but the survey does not show that they are associated with strong acceptance of the proposed price.

Five of the 38 respondents in this group already bought milk through community stores rather than supermarkets. This suggests some familiarity with alternative retail channels, but it does not tell us whether they would choose home delivery.

C.3.3 Price driven buyers (n = 19, 20%)

This group had the lowest willingness to pay across the three market segments. None of the 19 respondents stated a WTP of €2.00/L or more, and only three (16%) reached €1.50/L. More than one quarter of this group also report consuming 3–5 litres per week. Higher consumption would make a price increase more expensive in absolute terms, but the survey does not establish that consumption volume is the reason for their lower WTP. Based on the current survey, this group is unlikely to be a priority for a product priced around €2.10/L. However, the 19 respondents represent 20% of this sample, not 20% of the Lugo population, so the result should not be interpreted as showing that one-fifth of the Lugo market is unreachable.

Table 5. Behavioural profile of the two primary target segments

Behavioural variable	Quality & taste seekers (n=31)	Local & eco advocates (n=38)
Weekly milk consumption	1–3 L: 52%; <1 L: 32%; 3–5 L: 10%	1–3 L: 50%; <1 L: 32%; 3–5 L: 13%
Primary purchase channel	Supermercado: 90%; Tenda de barrio: 3%	Supermercado: 87%; Tenda de barrio: 13%
WTP ≥ €1.50	42%	21%
WTP ≥ €2.00	6%	3%
Brand repertoire examples	Pascual, CLA, Larsa, Feiraco, Deleite	Larsa, Feiraco, LEYMA, Deleite, Leche Rio, Xanceda
Alternative-channel use	1 already uses home-delivery	5 use neighbourhood shops

Source: Lugo consumer survey, May–June 2025 (Omar). Alternative-channel use refers to respondents already purchasing outside mainstream supermarkets, interpreted as evidence of existing willingness to deviate from the dominant supermarket habit.

C.4 Sociodemographic context

Sociodemographic information is relevant here because the feasibility of As Leiteiras depends not only on stated willingness to pay, but also on whether the likely buyers have the income, household structure, and channel access characteristics needed to support repeated home-delivery. Income affects the ability to absorb a €2.10/L price point; household size affects expected litres per stop and therefore route economics; age and digital familiarity affect the likelihood of adopting online ordering or subscription delivery; and place of residence affects customer density and route feasibility. For this reason, sociodemographic variables would normally be used to define the reachable consumer more precisely and to distinguish between general interest in local milk and a commercially viable customer segment. The survey did not collect sociodemographic data; age, gender, income level and occupation were not included in the questionnaire. The observations below draw on INE population data for Lugo as contextual background rather than as a substitute for respondent level data, and should be read as indicative rather than empirically grounded.

C.4.1 Age

Lugo has an ageing population, which is consistent with the wider demographic pattern in Galicia. The INE 2024 padrón records 99,482 inhabitants in the city, including a substantial older population. However, the survey itself did not record respondents' ages, so the effect of age on demand cannot be tested directly. Barreiro recalled that some older customers brought their grandchildren to the Santiago milk dispensers because the experience reminded them of how milk had been collected in the past, perhaps reflecting a degree of nostalgia for some customers in that project. However, there is no evidence from the Lugo survey that older consumers would respond in the same way to As Leiteiras. Age could also affect the use of online ordering, but this cannot be examined with the available survey data. The same problem applies to income: no respondent-level income information was collected.

No age profile can be assigned to the quality and taste segment because age was not included in the questionnaire. Any relationship between age and interest in the product would therefore need to be tested separately.

C.4.2 Household structure and consumption volume

The survey records consumption for individual respondents, while household size was not collected. Household demand therefore cannot be calculated reliably from these responses. Most respondents reported drinking 1–3 L per week, but this does not show how many litres a household would order in one delivery. The volume generated by each B2C delivery stop would likely be small as a result. The survey only records consumption for individual respondents, so it cannot be used to estimate litres per B2C household or per delivery stop. The financial model therefore uses 3 L/week per household as a separate planning assumption rather than as a value taken from the survey. Even under that assumption, B2C volume alone is unlikely to make residential delivery economically efficient. This supports giving greater priority to B2B accounts, where larger order volumes could improve route economics.

C.4.3 Income and purchasing power

No income data were collected in the survey. As a general reference, net income per person in Lugo province was €14,316 in 2023, compared with €15,036 for Spain and €15,319 in A Coruña province (INE, 2023). This lower income level is consistent with the price sensitivity found in the WTP results. It may also mean that the market for premium milk is smaller than in wealthier areas. At €2.10/L, As Leiteiras would cost about 2.3 times the Mercadona own-label price of €0.90/L, which is likely to make B2C adoption more difficult.

C.5 Synthesis: the reachable consumer

By taking into account behavioural, psychographic, and sociodemographic evidence, we can initially construct a profile of early adopters:

- Behavioural: Currently consumes 1-3 litres of milk per week; has purchased at least one Galician dairy brand; some respondents already purchase outside mainstream supermarkets.
- Psychographic: The survey points mainly to consumers who give more importance to quality or taste than to price. This group also shows a higher stated WTP than the local/eco and price-driven groups. However, only a small share reaches the €2.00/L bracket.

- Sociodemographic: No sociodemographic target can be defined from the survey because age, household size and income were not collected. These characteristics would need to be examined in future customer research.

The survey therefore identifies a relatively narrow group with characteristics that may be relevant to B2C testing. Applying the €2.00 WTP threshold and the 20% planning assumption to the Lugo population gives a proxy of approximately 637 demand units. The population-based figure gives only a rough indication of possible demand. It is not the customer number used in the financial model, which assumes 180 active B2C households in the stable year. Even with that assumption, household sales do not provide enough volume to cover the current operating costs. Appendix E therefore looks separately at B2B customers, where each account can contribute a much larger weekly volume.

C.6 Limitations

- This survey did not collect sociodemographic variables such as age, sex, family size, income or education level. As a result, the sociodemographic portion of this analysis relies on inference rather than direct empirical evidence. A follow-up survey or secondary data source (e.g. INE census micro-data for Lugo) would be needed to validate the inferences. The author had access to the raw response data (including question wording, which is how the “over 5 L/day” unit design issue noted above was identified), but not to the survey’s distribution and recruitment records, which precludes a full assessment of sampling frame coverage and non-response bias. Because the survey's distribution and recruitment records are unavailable, the direction and magnitude of any sampling bias cannot be quantified. The convenience sample may over-represent respondents closer to the founder's network or more comfortable with online surveys, but this cannot be confirmed from the available records.

- Convenience sample. The survey was distributed by the project founder through personal networks. No age data was collected, so the respondent pool's age skew cannot be quantified; a founder-network sample is plausibly skewed toward higher digital literacy and social proximity to the founder.
- Open text purchase driver question. Segment assignment is based on the analyst's classification of open text responses. Responses citing multiple factors (e.g. "price and local product") are assigned to a single primary category, which introduces subjectivity. A follow up structured question with forced single choice or ranked ordering would allow more reliable segmentation.
- No attitudinal or lifestyle questions. True psychographic profiling typically uses validated scales (e.g. environmental concern, health consciousness, food involvement) rather than purchase driver proxies. The psychographic segments identified here are exploratory and should not be treated as formally validated consumer archetypes.

Appendix D: Willingness to Pay — Extended Analysis

This appendix contains the WTP-by-consumption-volume and WTP-by-purchase-driver cross-tabulations (Tables 7 and 8), Figure 2 (WTP distribution chart), the full stated-preference instrument-design discussion, and the detailed regression interpretation and model diagnostics underlying the compressed treatment in Section 4.4. It has been relocated here to keep the main text within the required page limit. Original table and figure numbering is retained.

The willingness to pay (WTP) data reported here derive from a closed answer survey question administered to 94 residents of the Lugo area by the project founder (Omar) in May–June 2025. The samples were collected using convenience sampling or snowball sampling methods, and are essentially exploratory studies, lacking statistical representativeness and unable to represent the total population of Lugo. All WTP data in this section are stated WTP; research indicates that stated WTP overestimates actual purchasing behaviour. If this distinction has a substantial impact on the analysis results, it will be explicitly stated.

D.1 WTP distribution

The questionnaire used four price brackets, from less than €1.00 to €2.00–€2.50, plus an unanswered option. Table 6 reports the full distribution. Most respondents selected €1.00–€1.50/L (57.4%), while 26.6% were willing to pay at least €1.50/L. Only three respondents (3.2%) selected the highest bracket.

This highest bracket is the only one that includes the proposed B2C price of €2.10/L. Since the questionnaire did not ask about €2.10 directly, the three responses cannot be treated as confirmed acceptance of that price. They only show that €2.10 might be acceptable to a very small part of the sample. Omar reported that Sen Máis sells at around €2.10/L in Lugo, although this price has not been independently verified. Figure 2 shows the same WTP distribution graphically.

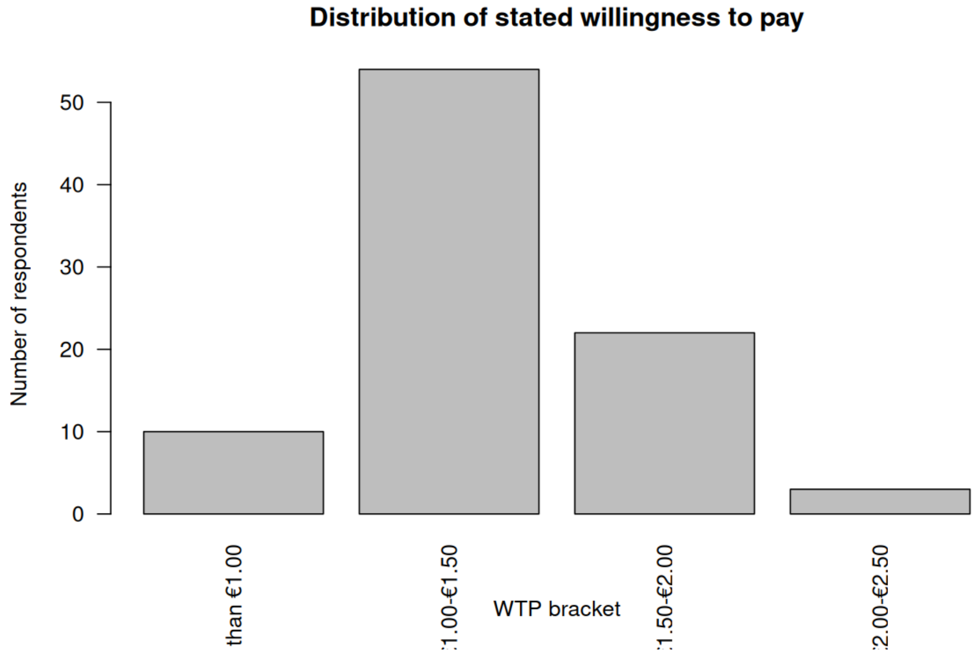

Figure 2. Distribution of stated willingness to pay (n = 89, excluding no answer respondents). Source: Lugo consumer survey, May–June 2025 (Omar).

D.2 WTP by consumption volume and purchase driver

D.2.1 Consumption volume

Table 7 shows that higher milk consumption was not associated with higher WTP in this sample. Among respondents drinking 3–5 litres per week, none selected €2.00/L or more and only 8% reached €1.50/L. The opposite pattern appears among those drinking less than 1 litre per week, where 41% were willing to pay at least €1.50/L.

Table 7. WTP by weekly consumption volume

Weekly consumption	n	WTP ≥ €2.00 (n / %)	WTP ≥ €1.50 (n / %)	Interpretation
Less than 1 L/week	27	0 (0%)	11 (41%)	Low volume, but relatively open to premium
1–3 L/week	45	2 (4%)	10 (22%)	Core segment; moderate premium acceptance
3–5 L/week	13	0 (0%)	1 (8%)	High volume, strong price resistance
More than 5 L/day	3	1 (33%)	1 (33%)	Very small n; not generalisable

Source: Lugo consumer survey, carried out by the project founder in May–June 2025. The "More than 5 L/day" wording issue is discussed in the note under Table 3.

The survey does not explain why this pattern occurs. It only shows that, within this sample, respondents with higher weekly milk consumption were less willing to accept the higher price ranges. Low volume buyers, by contrast, may be treating the product as an occasional or premium purchase, making them more tolerant of a higher price.

D.2.2 Primary purchase driver

Table 8 examines WTP against the primary factor respondents cited in their dairy purchasing decisions. Quality and taste oriented buyers show the highest premium acceptance, with 40% and 45% respectively willing to pay €1.50 or more. The "local product" segment is the largest single group, with 30 respondents, but only 23% are willing to pay €1.50 or more, and just 3% are willing to pay €2.00 or more.

Table 8. WTP by primary purchase driver

Primary purchase driver	n	WTP ≥ €2.00 (n / %)	WTP ≥ €1.50 (n / %)	Implication for positioning
Quality (Calidade)	20	1 (5%)	8 (40%)	Strongest premium segment
Taste (Sabor)	11	1 (9%)	5 (45%)	Highest WTP ≥ €1.50 rate; small n
Local product	30	1 (3%)	7 (23%)	Largest group; modest premium tolerance
Price (Prezo)	19	0 (0%)	3 (16%)	Lowest premium acceptance
Animal welfare	6	0 (0%)	1 (17%)	Small group; limited willingness
Organic (Ecolóxico)	2	0 (0%)	0 (0%)	Too small to draw conclusions

Source: Lugo consumer survey, carried out by the project founder in May–June 2025. Primary drivers were classified by the first or most prominent term in each open-text response; where a respondent listed multiple factors, the one judged most important was selected instead of automatically using the first, which involved researcher judgement rather than a fixed coding rule. The Ecolóxico category (n = 2) is included in the table for completeness, but the sample is too small to support any interpretation.

This discovery has a direct impact on market positioning. A strategy centred on "local product" messaging might seem natural given Sen Máis's Lugo area provenance, but it reaches a segment with limited price tolerance despite being the largest group of interested consumers. A positioning centred on quality and sensory experience (taste, freshness, whole-milk texture) reaches a smaller group with higher stated WTP. In practice, these messages need not be mutually exclusive, but the data suggest that "local" alone is insufficient to justify the premium price.

D.3 Stated WTP and the gap to real purchasing behaviour

Stated WTP in surveys consistently overstates actual purchasing behaviour for several reasons.

Hypothetical bias. As discussed in Section 2.1, stated WTP is well known to overstate real purchase behaviour. The Lugo survey used a simple closed-answer format without a simulated purchase decision, which is known to amplify this bias.

A note on instrument design. The Lugo survey did not include a “would actually purchase” validation question, nor a cheap-talk script of the kind recommended in the stated preference literature to alert respondents to the tendency to over-report willingness to pay (Cummings & Taylor, 1999). The 20% conversion rate applied in the demand model (Section 4.4) is a deliberately conservative planning assumption, not an empirically derived figure. Alternative elicitation methods (choice experiments, Becker–DeGroot–Marschak auctions, or real purchase pilots) would produce more behaviourally grounded estimates. In the absence of such data, the conversion-adjusted demand figures should be read as illustrative of the structural problem rather than as forecasts.

Social desirability. Questions framed around “local,” “organic,” and “ecological” products tend to elicit aspirational responses. For example, a respondent who normally buys lower-priced supermarket milk may still state a higher WTP for an organic product in a survey. This does not necessarily mean that the respondent would change their actual purchase.

Behaviour attitude gap. The survey also shows a difference between current purchasing behaviour and stated WTP. Of the 83 respondents who buy milk from supermarkets, only one (1.2%) stated a WTP of €2.00/L or more. Among the six respondents using neighbourhood shops, only one reached this level. High stated WTP is therefore uncommon in both groups.

For Lugo city, I use the 20% conversion assumption together with the WTP result to construct an illustrative B2C demand proxy. As noted above, the 20% figure is a planning assumption rather than a result of the survey. The calculation is:

99,482 (Lugo city population, INE 2024) × 3.2% (stated WTP ≥ €2.00) × 20% (conversion) ≈ 637 potential demand units (population-based proxy)

The 637 figure is a population-based proxy and should not be read as a customer forecast. The financial model separately assumes 180 active B2C households at 3 L/week. Under this assumption, B2C sales still do not cover the operating cost base, so additional B2B volume is required in the model.

Including the Tier 1 municipalities increases the estimate only modestly, from about 637 demand units in Lugo city to 718 across a population of 112,238. This is a relatively small increase, suggesting that B2C expansion into Tier 1 would have limited impact on overall demand. Any suburban expansion would therefore depend mainly on sufficient B2B demand.

D.4 Regression analysis of willingness to pay

An ordered logistic regression was used to examine which respondent characteristics were linked to higher WTP. This method was chosen because the four WTP categories have a clear order, but the distance between the categories is not equal. OLS was therefore not suitable. Ordered-choice models are the standard treatment for outcomes of this kind (Greene & Hensher, 2010). A multinomial logit model was also considered, but only three respondents were in the highest WTP category, so the estimates would have been unstable. Price-driven buyers were used as the reference group. Quality and taste were combined into one category, and eco and animal welfare were combined into another.

D.4.1 The impact of consumption volume, purchase channel, and purchase drivers on WTP

Several patterns are apparent, although the small sample size (n = 84) and convenience sampling mean the results are best read as directional rather than precise estimates. Among the purchase-driver variables, quality and taste show the strongest association with higher WTP, with an odds ratio of approximately 8.9. Compared with price-driven respondents, the quality and taste group has higher odds of being in a higher WTP category (OR ≈ 8.9). The same pattern appears for eco/welfare (OR ≈ 7.4) and local product (OR ≈ 4.6), although the estimated association is smaller for local product. Purchase channel is also important: respondents already buying through an alternative purchase channel show an odds ratio of approximately 9.5, suggesting that existing channel behaviour is at least as informative as stated motivation. People who are currently drinking milk (compared to those who do not drink milk) showed a greater impact, with an odds ratio of approximately 10.0. Among the predictors, current milk consumption has one of the larger coefficients, although the small "other" category is even larger and less stable. Current milk drinkers tend to report higher WTP, while heavier consumption is associated with lower WTP. For consumption volume, the odds ratio is about 0.34 for each step up in the consumption category, which is in line with the pattern shown in Table 7. These estimates need to be read cautiously because the model uses seven predictors with only 84 observations, and some predictors may be correlated. The coefficients are therefore useful mainly for identifying broad patterns rather than for precise prediction.

The survey gives a clearer starting point for further B2C testing: consumers who place less emphasis on price and are already comfortable buying outside the main supermarket channel. Existing Sen Máis or Arqueixal customers may also be worth approaching because they already know similar products. What remains unknown is whether familiarity with the product would be enough to make them choose home delivery. This would need to be tested directly, together with practical issues such as delivery times, minimum orders and bottle returns.

D.5 Implications for pricing strategy

The WTP results place several limits on the pricing options for As Leiteiras.

- The model keeps €2.10/L as the working B2C price, although the survey provides little evidence that many consumers would accept it. Omar reported that Sen Máis milk sells for roughly the same price in Lugo shops, but this still needs confirmation from the producer. Price is not the only problem. At the stable-year B2C volume used in the model, sales remain too low to cover the operating costs. For this reason, €2.10/L remains the Base Case price, while €1.90, €2.30 and €2.50/L are tested separately in the sensitivity analysis (Appendix E, Table 14).
- A lower B2C price may attract more customers, but the room for reducing the price is limited because the milk already costs €1.65/L. Although 26.6% of respondents stated a WTP of at least €1.50/L, this price would not even cover the purchase cost of the milk. A price above €1.65/L would create a positive gross margin on the milk itself, but this margin would still have to cover delivery and the other operating costs. The survey cannot show demand at €1.80 or €1.90/L because respondents selected price ranges rather than exact prices. What the brackets do show is a steep fall between the two levels that were measured: 26.6% at or above €1.50/L against 3.2% at €2.00–2.50/L. A price test across this range would be needed before assuming that any point within it is commercially workable.
- The B2B channel is considered separately because the purchasing decision differs from household WTP and the model assumes much higher weekly volume per account. However, no Lugo HORECA survey was conducted, so the proposed

€1.95/L B2B price and the value placed on organic, local or glass-bottle attributes remain unverified.

D.6 A note on non milk drinkers

Eight survey respondents indicated that they do not currently consume cow's milk (9% of the sample). Of these, 7 nonetheless provided a WTP response: 6 placed themselves in the €1.00–€1.50 bracket and 1 in the €1.50–€2.00 bracket. Although this finding cannot be generalised due to the small sample size, it indicates that the product may also appeal to some consumers who do not currently buy milk. Features such as glass bottles, home delivery, and local origin may make the product attractive to consumers who are open to a more premium alternative. This segment is not included in the financial scenarios because of the limited number of observations, but it could be explored further in future research.

Appendix E: B2B / HORECA Channel — Extended Analysis

This appendix provides the detailed material supporting Section 4.7, including the route-capacity comparison (Table 11), the Santiago de Compostela case and Barreiro interview, the B2C price-sensitivity analysis (Table 14), the decision criteria, client-acquisition discussion, and full list of limitations.

This section does not use the 94-response Lugo consumer survey because that survey covered household consumers only. Instead, the analysis draws on three sources: Omar's project notes and worksheet responses from May 2025, an interview with Ramón Barreiro, co-founder of a similar milk-delivery project in Santiago de Compostela (2010–2016), and illustrative scenarios based on the unit economics of the operating model. No separate HORECA survey was carried out. The section should therefore be treated as a feasibility assessment rather than direct market evidence.

E.1 Why B2B matters more than B2C to this model

Omar’s project notes also support this view. As Leiteiras was originally conceived mainly as a household delivery service, but the financial model does not support a B2C-only launch at the volumes tested. To build enough sales volume, the business would have to rely heavily on cafés, restaurants or other B2B accounts.

The driver alone represents a substantial operating cost. Based on the 2026 Lugo wage agreement for delivery drivers, the monthly cost is about €2,270 including employer social security contributions (Convenio Colectivo de Transporte de Mercancías por Carretera de la Provincia de Lugo, 2026). This replaces the earlier A Coruña benchmark used in the model. Spread over 20 to 22 delivery days, the driver costs roughly €103 to €113 per operating day. Milk sales must generate enough margin to cover this labour cost before the other delivery costs are considered.

The van adds another fixed commitment to the launch model. The Base Case uses a second-hand refrigerated vehicle priced at €18,925, with an eight-year useful life and a 17.5% residual value. Depreciation does not affect EBITDA, but it is included later in EBIT, net profit and NPV.

B2C earns more per litre, but B2B can deliver more volume at each stop. After VAT, the B2C margin is about €0.43/L at a selling price of €2.10/L and a milk cost of €1.65/L. For B2B, the corresponding margin is about €0.29/L at €1.95/L. The difference in route economics therefore depends heavily on litres delivered per stop. Table 11 shows daily contribution of about €86.54 at 10 L per B2B stop, compared with €88.27 in the illustrative B2C route. At 20 L and 30 L, the B2B figures increase to €173.08 and €259.62. The main assumption of 12.5 L per B2B stop produces about €108.17 per day, so the advantage over the B2C example is still fairly small at that volume.

The number of stops that can be completed in a day is another practical constraint. Omar reported that drivers he consulted regarded about 80 household stops as a reasonable daily reference, while around 30 stops was considered more realistic for bars and cafés because each delivery would normally be larger. These are informal estimates rather than industry benchmarks. For B2C, Table 11 uses 68 stops per day rather than 80 because the working day also includes loading and the trip to Sen Máis. The model allocates about 80 minutes to this collection trip before customer deliveries begin. The detailed time and cost assumptions are set out in the Break-even Analysis and Operating Costs sections of the financial model.

Table 11. Illustrative route-capacity comparison by channel type, per delivery day

Channel	Stops/day	Litres/stop	Litres/day	Gross margin/day
B2C illustrative route (3 L/stop)	68	3	204	€88.27
B2B only: small accounts (10 L/stop)	30	10	300	€86.54
B2B only: medium accounts (20 L/stop)	30	20	600	€173.08
B2B only: large accounts (30 L/stop)	30	30	900	€259.62
B2B illustrative route (12.5 L/stop)	30	12.5	375	€108.17

B2C price €2.10/L, B2B price €1.95/L, milk purchase price €1.65/L. Driver daily cost: approximately €103–€113. Figures cover gross margin on milk only; other operating costs are excluded. These rows illustrate route capacity at different stop intensities on a given day; they are not the stable-year sales volumes used in Table 12, which are lower because not every customer is delivered to every day.

A purely B2C operation at the model's actual stable-year assumption of 3 litres per stop generates approximately €88.27 in gross margin at 68 stops per day. That is roughly 78–86% of the driver's scheduled daily cost, still short of covering it, and before fuel and other operating costs are deducted. A B2B route at the illustrative 12.5 litres per stop generates approximately €108.17 at 30 stops per day, close to but not reliably above the driver's daily cost on its own. A B2B route with medium sized accounts at 20 litres per stop generates €173.08, roughly 1.5–1.7 times the driver's daily cost. Even a basic B2B route barely covers the driver's daily cost before fuel and other expenses are added, so the route cannot rely on B2C volume alone: B2B accounts have to carry most of the delivery litres for the numbers to work at all.

E.2 Evidence from the Santiago de Compostela case

The Santiago project was much more vertically integrated than the model proposed for As Leiteiras. The cooperative kept the cows, processed the milk, operated the vending machines and organised distribution itself. Barreiro recalled that the project eventually had eight machines and sold roughly 1,500–3,000 litres per day in Santiago and nearby areas. As Leiteiras would not reproduce this structure, since the milk would already be processed and bottled by Sen Máis. Its role would mainly be collection, delivery and customer management. The Santiago volumes are therefore useful as historical context, but they are not a realistic sales benchmark for Lugo.

Barreiro also linked the founders' decision to leave the project to the amount of work involved. They were handling production, processing, packaging, sales and delivery at the same time. When the business required further investment, they decided not to continue and sold most of the machines. This experience is relevant to As Leiteiras mainly as a warning against taking on too many activities within a small operation. This experience is one reason why the proposed As Leiteiras model is narrower. It focuses on delivery rather than trying to manage the whole dairy production process. Sen Máis would be responsible for production, processing and bottling. As Leiteiras would therefore avoid having to invest in and operate these parts of the business itself.

The Santiago case shows that bars, restaurants and cafés were part of the customer base. However, the case does not show how much of the total sales came from HORECA customers. Barreiro identified packaging, price, sanitary controls and delivery as important points in the former project. The interview notes do not provide a separate ranking of the factors that attracted HORECA customers. The dispensing machines also gave the Santiago project a visible physical presence. As Leiteiras would not have the same type of sales point in Lugo. Direct contact with potential B2B customers may therefore be more important for the proposed model.

HORECA and household customers face different purchasing decisions. A café buys milk as an input for its own products, while a household pays the full retail price directly. For this reason, the household WTP results cannot be applied to HORECA customers in the same way.

Barreiro reported that his five current restaurants in Santiago use Sen Máis milk. This information comes from the interview and has not been independently verified. The Barreiro interview confirms that Sen Máis supplied five restaurants in Santiago, but it does not explain why those businesses chose the product. The 25 L/week used for each B2B account in the financial model comes from a separate planning assumption, not from the interview or a Lugo HORECA survey. Under the model assumptions, a B2B account contributes substantially more weekly volume than a B2C household.

The five Santiago restaurants are useful as a real example of Sen Máis being sold through the HORECA channel. They do not provide enough evidence to estimate how common this type of demand is elsewhere in Galicia.

E.3 Viability scenarios

Because no primary HORECA survey was carried out in Lugo, this section works with scenarios rather than a single estimated B2B demand forecast. Each scenario keeps the stable-year assumptions from the financial model, including a fixed weekly consumption per customer, while varying the number of B2C households and B2B accounts; the Conservative case additionally lowers the B2B price.

In the stable-year Base Case, the model assumes 180 B2C households at 3 L/week and 30 B2B accounts at 25 L/week. Together, these assumptions give annual sales of 67,080 litres. B2B accounts make up only about 14% of the total accounts in this scenario, yet they generate 58% of annual volume, so household sales alone do not carry the model: the 30 B2B accounts deliver more litres than the 180 B2C households do. Whether this extra volume is enough to cover costs is tested in the scenarios that follow.

E.4 Combined B2C and B2B sensitivity analysis

The sensitivity analysis keeps the main Base Case assumptions from the financial model: €2.10/L for B2C, €1.65/L milk purchase cost, 3 L/week per B2C household and 25 L/week per B2B account. The Base and Upside cases use a B2B price of €1.95/L. The Conservative case uses €1.85/L. B2C demand pool: 99,482 (Lugo city, INE 2024) × 3.2% (stated WTP ≥€2.00) × 20% conversion = 637 population based demand proxy. The stable-year target of 180 B2C customers refers to households, while the 637 estimate refers to population-based demand units. The two figures are therefore not directly comparable. The 180-household target is included only as a rough reference.

Omar identified 272 possible B2B establishments in Lugo. This figure has not yet been independently verified and includes both hospitality businesses and some specialty retailers. I therefore use it only as a B2B prospect pool, rather than as an estimate of the total HORECA market.

Table 14. B2C price sensitivity at Base Case volumes

B2C price	B2C customers	B2B accounts	Annual litres	Operating costs/year	Revenue/year (net of VAT)	EBITDA/year
€1.90/L	180	30	67,080	€160,981	€124,425	−€36,556

B2C price	B2C customers	B2B accounts	Annual litres	Operating costs/year	Revenue/year (net of VAT)	EBITDA/year
€2.10/L (Base)	180	30	67,080	€161,359	€129,825	−€31,534
€2.30/L	180	30	67,080	€161,737	€135,225	−€26,512
€2.50/L	180	30	67,080	€162,115	€140,625	−€21,490

Table 13 changes customer numbers, while Table 14 keeps customer numbers fixed and changes the B2C price. The price sensitivity test holds customer numbers at 180 B2C households and 30 B2B accounts. With sales volume unchanged, most operating costs remain the same when the B2C price changes. Marketing is an exception because the model calculates it as 7% of revenue. For this reason, each additional €1 of net revenue adds roughly €0.93 to EBITDA.

Increasing customer numbers improves the result, but not enough to remove the loss in either upside case. An additional 15 B2B accounts raises annual EBITDA by about €2,700, while adding 60 B2C households produces a similar improvement. The B2B case achieves this with fewer additional accounts because each one contributes more weekly volume. Even so, the increase tested here is still below the level required for positive EBITDA.

Under the current assumptions, profitability would require a larger change than the upside cases tested here, either in sales, pricing or costs. In the Base Case, 180 B2C households and 30 B2B accounts produce 67,080 L of annual sales and €129,825 of revenue net of VAT. EBITDA is still approximately −€31,534.

E.5 Decision criteria and client acquisition logic

B2B and B2C customers need to be considered separately because their purchasing patterns and order volumes are different. The difference is clearer when volume is compared per account. The financial model assumes 25 L/week for each B2B account. This is a planning assumption and has not been validated through a Lugo HORECA survey. B2B and B2C volumes are different in the financial model. The model assumes 25 L/week per B2B account and 3 L/week per B2C household. The consumer survey cannot independently confirm the household figure because household size was not collected. The larger assumed volume per B2B account is one reason why this channel is important to the route economics.

Decision criteria. For household consumers, the Lugo survey shows that price is the main factor listed by around 21% of respondents. It also shows that only around 3% of the sample is willing to pay more than €2.00/L. B2B buyers may make the purchase decision differently. For example, a restaurant or café owner considers the cost of the milk as part of the total cost of the products offered to customers. B2B buyers may evaluate the purchase differently because milk is an input into products sold to customers. However, no Lugo HORECA research has tested whether buyers value organic production, local origin or glass packaging, or whether the proposed €1.95/L price is acceptable.

Payment and logistics terms. The financial model assumes a 30-day payment period for B2B customers. This is a model assumption, not a payment term observed from Lugo businesses. With a 30-day payment period, the estimated B2B receivables are around €6,300 in the stable year. This amount is not currently included in the cash-flow forecast, so the model may understate the working-capital requirement.

The model also assumes that HORECA orders are more regular than B2C orders. In the Base Case, B2B accounts are assumed to receive deliveries twice per week at scheduled times. No Lugo HORECA data were collected to confirm this pattern. If this assumption is realistic, regular HORECA orders could make delivery routes more efficient.

E.6 Limitations and Decision Implications

B2B analysis still relies on some assumptions that have not been independently verified. The main uncertainties are summarized below.

- Sen Máis supply capacity. The model currently uses 500 L/day as the available supply from Sen Máis, based on information provided by Rosa at the Lugo store in May 2026. This figure still needs confirmation from the producer. It is also well below the roughly 965 L/day required for the lower-bound full-cost EBITDA break-even point. If 500 L/day is close to the actual limit, the business would run into a supply constraint before reaching break-even volume.
- B2B prospect universe in Lugo. Omar identified 272 possible B2B customers from a local hostelería listing: 80 cafés and bakeries, 140 restaurants, 25 hotels and 27 specialty food shops. Because the list includes retail businesses as well as hospitality establishments, I use it as a prospect pool rather than as a measure of the Lugo HORECA market. The total has not been independently checked and could later be compared with INE DIRCE or IGE business data.
- Realistic acquisition rate. The 272 establishments indicate the size of the possible B2B pool, but not how many would actually become customers. No interviews or survey have been carried out with Lugo HORECA businesses, so I do not use the Santiago case to estimate a conversion rate. It only shows that restaurants and cafés can be a market for this type of milk.
- Sen Máis's existing HORECA relationships. Barreiro reported that five of his restaurants in Santiago use Sen Máis. The available material does not show how many cafés, restaurants or other business customers Sen Máis currently supplies in Lugo. It is therefore not possible to tell how much of the potential Lugo B2B market is already being served by the supplier.

- B2B pricing tolerance. The B2B price of €1.95 per litre is a working assumption based on the price difference between farm delivery price and retail price. No HORECA buyer has been asked whether this price point is acceptable relative to their current milk sourcing costs.
- Segmentation within the HORECA universe. Omar's worksheet responses suggest that the 272 B2B establishments should not all be treated as equally relevant targets. He distinguishes between businesses that compete on quality or eco positioning and those that compete mainly on price. In his view, the first group is more likely to be suitable for As Leiteiras. This distinction has not yet been reflected in the 30-account stable-year target or in the financial model's B2B sizing. A future version could apply a similar quality/local/price segmentation to the B2B market.
- Rural accommodation / hotel channel resistance. Omar's worksheet responses describe a preliminary conversation with a rural guesthouse about branding home-delivered bottles as a "good morning" gesture placed at each guest's door. The owner declined, because the property's business model depends on prompting guests to check out promptly rather than encouraging them to linger. This is the only direct qualitative evidence gathered so far on hotel/rural-accommodation demand, and it points to a specific source of resistance within the "Hotels (small/rural)" segment that is not currently reflected in the financial model's 4% penetration assumption for that segment.

Primary research on these questions would sharpen these estimates, especially a small number of structured conversations with café or restaurant owners in Lugo. It is not a precondition for the recommendation below. In their absence, the scenarios in Table 13 should be treated as illustrative of the commercial logic rather than as demand projections.

These remaining market and operational uncertainties do not prevent a decision on the proposed full-cost launch. The Base Case and all tested alternative scenarios remain EBITDA-negative under the current price and cost structure. The recommendation is therefore a no-go for purchasing a dedicated refrigerated van and hiring a full-time driver. The unresolved items represent additional downside risks rather than conditions that could reasonably reverse the present conclusion. A future pilot or a different operating model would need a separate financial analysis. The current study does not provide evidence that either option would be profitable.

Appendix F: Marketing Action Framework

Given how little room the results leave for a conventional market launch, this appendix sets out what the analysis does support: which customers to approach first, how to position the offer, and which channel to prioritise if the project goes ahead.

F.1 Target Segment

The clearest B2C target is quality- and taste-driven milk drinkers: 42% of them would pay at least €1.50/L, against 21% for the local/eco group, and the regression gives this driver the strongest effect of any tested (OR ≈ 8.9). No age, household size or income data exist for this group, so no sociodemographic profile is possible; anything beyond their stated purchase driver would be guessing. A more practical starting point is people who already buy Sen Máis, Xanceda or Arqueixal. They already know what premium milk costs and tastes like, though whether they would switch to a delivery service is not something the data can answer. The 637-unit figure from Section 4.4 belongs here too, but as a warning rather than a target: it is a population-based proxy, not a count of people who would buy.

F.2 Positioning

The survey gives stronger support to quality and taste than to local origin as a basis for positioning. Respondents were already familiar with Galician brands such as Larsa and Feiraco, which are available at much lower prices than the proposed €2.10/L. Local origin alone is therefore unlikely to explain such a large price difference. Freshness may offer another point of differentiation, although the questionnaire did not test it separately. In practice, As Leiteiras would be offering several features together: fresh organic milk, home delivery and returnable glass bottles. The survey cannot show how much value consumers attach to each feature individually. Local origin and organic certification can still support the overall offer, but the results do not show that either one alone would justify the proposed price.

F.3 Channel Priority

B2B contributes much more volume per account than B2C in the financial model. A B2B customer is assumed to buy 25 L/week, compared with 3 L/week for a B2C household, so one B2B account provides roughly the same weekly volume as eight households. With two deliveries per week, the average B2B drop is 12.5 L. The scenarios tested still remain EBITDA-negative, but the larger volume per stop means that fewer deliveries are needed to move the same quantity of milk.

For this reason, I would not commit to a dedicated driver and refrigerated van before testing B2B demand in Lugo directly. If enough B2B customers can be identified, household deliveries could then be added around those routes where practical. Existing Sen Máis customers in Lugo would be a sensible group to approach first because they already know the product and are familiar with its approximate retail price.

F.4 Validation Geography

Lugo city should remain the first area for any future market test. It contains about 88.6% of the combined Core and Tier 1 population and has by far the highest population density in the area. O Corgo and Guntín are less suitable for an initial route because neither lies on the Sen Máis collection route, so serving them would add extra distance rather than fit naturally into the daily pickup journey.

F.5 Validation Thresholds and Operating Targets

The figures below come from different parts of the analysis and should not all be read as short-term targets.

Before committing to a van or a full-time driver, two issues need direct evidence: how much Lugo HORECA businesses would actually buy, and what supply terms Sen Máis would be willing to offer. Any early test should focus on these questions rather than trying to reproduce the stable-year model. Interviews or a small commercial test could be used to check B2B interest, likely weekly volume and acceptable prices. B2C demand could be tested along the same routes where practical.

The stable-year model is a separate reference point. It assumes 30 B2B accounts at 25 L/week and 180 active B2C households, yet still produces EBITDA of about −€31,534. For future B2C research, existing Sen Máis buyers and consumers who already purchase outside mainstream supermarkets would be the most relevant groups to approach first.

Appendix G: Disclosure of Generative AI Use

Generative artificial intelligence was used as a supporting tool during selected stages of the preparation of this thesis. The main tool used was ChatGPT (OpenAI). Its use focused on organising information, discussing the research structure, identifying gaps in the available information, checking consistency between the financial model and the written analysis, and improving the clarity and concision of the document.

The underlying project information, primary data, survey responses, interview material and source documents were not generated by AI. These materials originated from the project promoter, the survey conducted in Lugo, interviews and project documents. The financial model was likewise built by the author from these sources. AI was used to review its internal consistency, identify possible issues and discuss potential revisions; any changes to assumptions, structure or calculations were verified and implemented by the author.

Date	Purpose of AI use	Description
17 April 2026	Research framing and development of the research questions	ChatGPT was used to discuss how the initial business concern could be developed into a suitable thesis problem. The discussion helped broaden the focus from the specific question of when it would make financial sense to hire a full-time delivery person towards a wider assessment of the market, operational and financial conditions under which the As Leiteiras home-delivery model could become viable. Possible sub-questions on demand, customer segments, delivery operations and financial viability were also discussed. The final research questions and scope were selected by the author.
29 April 2026	Organisation of information provided by Omar	ChatGPT was used to help organise and synthesise the different documents and background information provided by Omar at that stage of the project. The purpose was to group the material into relevant themes, distinguish information that could support the thesis from issues that still required clarification, and create a clearer basis for the subsequent research and financial analysis. The underlying project information came from Omar and the original documents, not from AI.

Date	Purpose of AI use	Description
11 May 2026	Identification of missing information and follow-up questions	ChatGPT was used to identify areas where the available information was still incomplete for the financial and operational analysis. Examples included pricing, payment terms, supply arrangements with Sen Máis, bottle handling, delivery operations and commercial conditions. It also helped organise possible follow-up questions to be checked with Omar or other relevant project contacts.
28 June 2026	Consistency check between the financial model and the thesis	ChatGPT was used as an additional checking tool to compare the Excel financial model with the written thesis and identify possible inconsistencies in assumptions, units, terminology, figures and conclusions. Any proposed corrections were reviewed against the spreadsheet calculations and the original project information before being incorporated.
19 August 2026	Editing, condensation and removal of repetition	At a late stage of the thesis, ChatGPT was used to discuss which parts of the document could be shortened or removed without changing the underlying analysis. The purpose was to reduce repetition, improve readability and keep the document concise. Decisions about what to delete, retain or rewrite were made by the author.

Nature and limitations of the AI assistance

ChatGPT was used primarily as a research-support and editing tool rather than as a source of evidence. Its functions included brainstorming and structuring the research problem, organising project information, identifying missing information, checking internal consistency, and supporting revisions for clarity and concision.

ChatGPT was not treated as an academic or factual source for substantive claims in the thesis. Where factual information, numerical assumptions, market information, survey results or interview evidence appears in the thesis, the relevant original source or project material was used instead. The author remained responsible for verification, analysis, interpretation and the final conclusions and recommendations.